

Human Capital and Mobility in the Executive Labor Market^{*}

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Abstract

We estimate a search model of executive careers to quantify the relative importance of human capital accumulation (both general and firm-specific), managerial bargaining power, and imperfect labor market competition in shaping compensation and mobility in the market for US corporate executives. The composition of human capital is career-dependent and varies widely across managers: over tenure, firm-specific capital is the greatest driver of wage growth, whereas over experience in the labor market, job search and competition dominate. Firm-specific capital can help explain the high rate of internal CEO hires and low observed cross-firm CEO mobility. We further show that labor market competition (relative to pure bargaining power) makes up the majority of realized CEO surplus capture, and that firm-specific human capital positively interacts with competition in determining CEOs' shares of rents as it raises the match-specific quality between the firm and manager.

Keywords: CEOs, the market for CEOs, executive pay, executive mobility, human capital accumulation, on-the-job search, structural estimation, firm-specific human capital

^{*}This paper is a work in progress and results are subject to change, all errors are our own.

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1. Introduction

In this paper, we quantify the relative importance of human capital accumulation, both general and firm-specific, managerial bargaining power, and imperfect labor market competition in determining compensation and mobility in the labor market US corporate executives. We do so by posing and estimating a structural model of the working life of the manager, and our paper contributes to the literature in three ways.

First, decomposing managerial human capital into its general and firm-specific components has important implications for the study of executives: prevailing theories rationalize the recent rise in CEO pay via a premium on general CEO skill (e.g., [Gabaix and Landier, 2008](#); [Murphy and Zabojnik, 2007](#)).¹ A feature of these models is that competition for generalist CEOs induces higher wages via an outside option channel; a direct empirical implication is that CEOs should move frequently across firms. Yet, as shown in [Cziraki and Jenter \(2024\)](#), around 72% of new CEOs at the largest companies in the US are internally promoted, and only a small percentage of external hires are poached CEOs ([Graham et al., 2020b](#); [Cziraki and Jenter, 2024](#)), suggesting that firm-specific managerial skill may be more important than previously considered.²

Second, understanding the contributions of pure bargaining power and imperfect labor market competition to managerial surplus capture informs the large literature on CEO bargaining power and influence in determining CEO compensation.³ In settings without an explicitly-modeled labor market, competition for CEO services that induces high CEO surplus capture may be incorrectly attributed to pure bargaining power. To the extent that agency frictions and CEO bargaining power are the same, agency frictions in CEO compensation may be overstated.

¹[Gabaix and Landier \(2008\)](#) and [Edmans et al. \(2009\)](#) are examples of competitive assignment models that allow for complementarity between CEO skill and firm size, which thus helps explain the rise in CEO pay. [Murphy and Zabojnik \(2007\)](#), in a separate but related setting, shows that general managerial skill improves managers' outside options, which can also help rationalize the increase in CEO pay.

²[Cziraki and Jenter \(2024\)](#) show that 73% of new CEOs at S&P500 firms are internal hires from 1994-2012, and a large proportion of external hires are known to the Board (either a former executive, or a Board member). [Graham et al. \(2020b\)](#) show that 68% of new CEOs are internal hires (a current or previous officer of the firm) for a fuller sample of NYSE/Amex firms from 1933-2011.

³See, e.g., [Core et al. \(1999\)](#); [Bertrand and Mullainathan \(2001\)](#); [Bebchuk and Fried \(2003\)](#); [Fahlenbrach \(2009\)](#); [Bebchuk et al. \(2011\)](#); [Morse et al. \(2011\)](#); [Coles et al. \(2014\)](#).

Third, we show that the managerial human capital and surplus capture channels interact in determining mobility and compensation. Firm-specific human capital accumulation increases the manager’s match quality with the firm over tenure, which can help explain the low rate of cross-firm mobility. However, it also has a positive impact on realized managerial surplus capture: more firm-specific skill raises the opportunity cost of losing the manager, making the firm more likely to match attractive outside offers. As such, firm-specific human capital can help rationalize simultaneously low mobility and high realized bargaining power of managers.

Our structural model is a comprehensive yet tractable characterization of executive careers, (both non-CEO and CEO), incorporating both general and firm-specific human capital accumulation; managerial bargaining power; labor market competition and mobility (including both internal and external promotion opportunities); and executive and firm heterogeneity. Unlike much of the existing literature which examines firm demand for executive talent, we instead model the executive’s career through the lens of job search (Bagger et al., 2014), in which these forces jointly determine mobility and compensation growth throughout a manager’s career.⁴

Managers in the model may be employed as a (non-CEO) executive or as the CEO of the firm.⁵ Over their careers, executives may receive job offers — to move horizontally, be promoted internally to CEO, or to be promoted externally to CEO; the arrival rates of these job offers differ and are estimated in the data. CEOs may receive job offers to be CEOs at other firms.

We follow an important strand of the search literature (Postel-Vinay and Robin, 2002; Cahuc et al., 2006; Bagger et al., 2014) and model wage contracts as piece-rate contracts: managers receive a portion $R \in [0, 1]$ of their contribution to firm output. When a manager receives an attractive outside offer, the incumbent and poaching firm may bargain over the executive’s services (in the spirit of Rubinstein, 1982). Firm-switching events occur when the poaching firm values the manager more than the incumbent. On-the-job search leads to stochastic, discrete in-

⁴Indeed, while firm-specific human capital can help explain preference for insiders and low CEO mobility (Cziraki and Jenter, 2024), its impact is not immediately separable from search frictions inhibiting the movements of CEOs across firms (such as a preference for internal CEO promotions He and Schroth, 2024). Moreover, differences in wages across executives and firms may be attributable to time-invariant (possibly unobserved) heterogeneity, either across managers or positions, or both.

⁵To limit confusion, throughout the paper we refer to non-CEO executives as “executives” and CEOs as “CEOs.”

creases in pay as firms bargaining over managerial services, even if the manager ultimately stays in their current position.

We adapt the setting in [Bagger et al. \(2014\)](#) to allow for firm-specific human capital (in addition to general) and for both external and internal promotion of executives. Importantly, we let search rates and executive bargaining power differ across internal and external promotion opportunities, which enables us to distinguish a preference for insiders ([He and Schroth, 2024](#)) from the impact of firm-specific human capital on internal vs. external executive mobility.

One insight from the model is that firm-specific human capital accumulation leads to increased match-specific productivity between the firm and manager over the manager's tenure. This makes the manager less likely to be tempted away by poaching offers as they advance at the incumbent (decreasing job-to-job transitions). However, firm-specific human capital increases the likelihood that the incumbent firm is willing to match attractive outside offers, precisely because of the increased match-specific productivity. We show theoretically that cross-firm mobility decreases with firm-specific human capital accumulation, whereas within-firm mobility (contract renegotiation due to outside offers) increases.

Further, our model admits closed-form solutions for managerial contracts, and we show theoretically that, all else equal, an internally promoted CEO receives a lower share of rents than externally-hired candidates, with poached CEOs receiving the greatest share, and this holds for any level of firm-specific human capital. This result stems from the relative outside options of the three types of manager. As such, our model predicts that CEOs receive path-dependent contracts, depending on whether they were internally or externally hired.

We estimate the model on a rich panel of executive careers spanning 1992-2023 (Execucomp), combined with manually-collected data on managers' tenures at firms and their experience in the executive labor market. This allows us to track experience, tenure, compensation and mobility over the working life of a manager. Transitions of executives (non-CEO and CEO) across firms, and the resultant impacts on wages and mobility, allow us to separately identify general and firm-specific components of executive human capital.

Our estimation produces several sets of results. First, we decompose managerial human capital into its general and firm-specific components. This is a standing open question in corporate governance and the answer has important consequences for the literature on executive compensation. We find that, on average, CEO human capital is $YY.Y\%$ general and $XX.X\%$ firm-specific. However, given that managers lose firm-specific capital in job-to-job (and job-to-unattachment) transitions, the proportions differ depending on the length of tenure; for example, firm-specific capital makes up roughly $XX.X\%$ of the total for managers in the first 10 years of tenure, whereas this proportion is about $XX.X\%$ across experience (the difference can be explained by job-to-job transitions early on in CEO careers). Similarly, firm-specific capital has a relative larger impact on compensation across tenure (accounting for about 45% of cumulative wage growth within tenure) than across labor market experience (about 25%).

We also analyze how firm-specific human capital impacts the hiring of internal vs. external CEOs. By simulating a counterfactual with no firm-specific skill and comparing to the baseline estimation, we find that removing firm-specific human capital lowers the proportion of internal CEO hires by about 15 percentage points (a decrease from 63% to 48%, or nearly a 24% fall), with about two thirds of the decrease going to poached CEOs (the remainder to poached executives). While search frictions and strategic cross-firm competition explain the majority of the observed rate of internal vs. external hiring, this result contributes to the understanding of mobility in the CEO labor market, stressing the importance of incorporating both firm-specific human capital and imperfect labor market competition ([Cziraki and Jenter, 2024](#)).

Our second set of results concerns realized managerial surplus capture and labor market competition. Our estimation allows us to decompose CEO rent sharing into pure CEO bargaining power and labor market competition pushing CEO wages up (which is inspired by [Cahuc et al., 2006](#)). We find that, on average, CEOs capture about 56% of rents, which is closely in line with estimates from papers in the literature. However, our estimate of pure CEO bargaining power is only 13.6%, suggesting that about 75% of realized CEO surplus capture is driven by labor market competition.

We also find that realized CEO surplus capture varies depending on if the CEO was internally promoted or externally poached, with poached CEOs commanding the greatest share of rents: internal CEOs capture 51.5% of surplus, poached CEOs 68.6%.⁶ These results contribute to the literature on agency frictions and bargaining power in executive compensation, showing that endogenizing labor market competition for executive services is crucial for understanding patterns in executive compensation.

Lastly, we show that firm-specific human capital impacts realized CEO surplus capture. We study how CEOs' shares of rents evolve over the tenure of the CEO in the baseline estimation and a simulated counterfactual without firm-specific capital, and show that firm-specific capital nearly doubles the CEO's share of rents. This stems from the theoretical result concerning how firm-specific human capital raises the firm's willingness to match attractive outside offers. Taken together, firm-specific human capital can help explain how market competition for CEO services can lead to low cross-firm mobility and high CEO surplus capture.

The paper is organized as follows. The rest of this section discusses our paper's place in the literature. Section 3 introduces the theoretical model. Section 4 discusses our estimation and identification strategies and Section 5 gives the estimated parameters and discusses model fit. Section 6 analyzes CEO bargaining power and labor market competition. Section ?? analyzes the role of firm-specific human capital (and other channels) in determining mobility and compensation. Finally, Section 8 concludes. Model proofs and additional details are in Appendix A, and estimation details are contained in B.

1.1. Literature Review

Our paper lies in the intersection of the on-the-job search literature in labor economics, and the study of executives in corporate finance. On the labor side, our paper is closely related to preceding work concerning on-the-job search, bargaining, and human capital. Starting with [Postel-Vinay and Robin \(2002\)](#) and [Cahuc et al. \(2006\)](#), the authors develop and microfound a workhorse

⁶This result stems from our theoretical result that poached CEOs receive strictly larger contracts than internally-hired CEOs.

structural model of careers with on-the-job search. [Bagger et al. \(2014\)](#) continues by adding general human capital accumulation to the model (among other features), and study its effect on worker careers.

We expand these models in several key ways. First, we allow the worker to accumulate both general and firm-specific capital, and show that firm-specific skill has previously unstudied impacts on labor market mobility and renegotiation over worker careers.⁷ Second, we allow the worker to possibly experience both internal and external mobility throughout their career, by allowing managers to be promoted to CEO at their current firm, as well as receive job offers to be CEO at other firms.

On the corporate finance side, the relative importance of general and firm-specific skill, or more specifically how this relative importance can explain CEO wage growth over time, has been extensively studied. Indeed, a large literature has arisen which stresses the importance of transferable executive skill in explaining the large observed increase in CEO wages over time.⁹ However, to our knowledge, no paper has explicitly attempted to directly quantify the weights of general and firm-specific human capital in managerial skill.

[Murphy and Zabochnik \(2007\)](#) (and the related [Murphy and Zabochnik, 2004](#)) model the firm's choice of an internal vs. an external candidate, and show that, in a market with an elastic supply of executive labor, a larger importance on general executive skill (relative to firm-specific) can rationalize the observed increases in executive pay that have been observed in the data. As [Custódio et al. \(2013\)](#) point out, a direct implication of a competitive executive labor market is that firm-specific human capital receives a lower premium in wages, because firm-specific human capital does not improve the executive's outside option.

However, as pointed out by [Cziraki and Jenter \(2024\)](#), the labor market for CEOs (especially

⁷As pointed out by [Lazear \(2009\)](#) and [Bagger et al. \(2014\)](#), firm-specific human capital has received relatively less attention than general in the labor economics literature, primarily because of the focus on rank-and file workers.⁸ However, firm-specific human capital is likely to be more important in the executive labor market, where, for example, fostering corporate culture ([Graham et al., 2022](#)), or efficiently deploying the firm's factors of production are important skills that may not transfer.

⁹This is a large, important recent literature in corporate finance, for example, [Gabaix and Landier \(2008\)](#); [Terviö \(2008\)](#); [Edmans et al. \(2009\)](#); [Murphy and Zabochnik \(2007\)](#); [Custódio et al. \(2013\)](#); [Frydman \(2019\)](#)

at large firms) seems to be highly frictional, with little movement of executives across firms as wages change and with executives rarely using their outside option; given these frictions, they suggest that the market for CEOs may involve firm-specific capital and frictions, in which the firm and CEO bargain to split the surplus generated by the match.

In a model of the executive labor market with on-the-job search, managerial bargaining power, in which incumbents and poachers compete for managerial talent, the above predictions of [Murphy and Zabojnik \(2007\)](#) and [Custódio et al. \(2013\)](#) are not necessarily true. While the manager’s utilization of the outside option may decrease with tenure, the importance of the *renegotiation option*, in which managers use outside offers to improve their incumbent contract, become relatively more important. Moreover, our estimates contribute to understanding of the role of labor market competition in determining CEO rent shares, which contributes to the structural literature on the executive labor market ([Taylor, 2013, 2010](#); [Barry, 2023](#); [Page, 2018](#)).

Several papers have measured the importance of general and firm-specific skill in high-skilled industries ([Gao et al., 2021](#); [Ma et al., 2023](#)). These papers allow the worker to accumulate human capital via learning-by-doing, and study how this shapes mobility choice across career. Our paper complements these studies by (i) focusing on the managerial labor market at US public companies more generally, and (ii) explicitly studying how firm-specific capital shapes mobility paths across careers. Indeed, our analysis on the role of job search in determining managerial rent shares is similar in flavor to the measure of “bargaining capital” in these papers.

2. Data

We estimate our model on a rich panel dataset covering executive careers at US public firms from 1992 to 2023. We supplement data from Execucomp with manually-collected data on when managers first joined their firm. Doing so allows us to accurately define managerial tenure and measure firm-specific human capital.

We classify each manager in our sample as being employed in an executive (non-CEO) or CEO position, or unattached. Unattachment refers to managers who stop working at a firm in

our sample for at least one year, but begin work at another public firm later in their career. We label managers who drop out of the sample permanently as retired.¹⁰

Our measure of compensation is Execucomp's TDC1, which includes each year's salary, bonus, stock grants, option grants (valued using Black-Scholes), long-term incentive payouts, and other non-equity incentive payouts. This measure of pay clearly contains (long-term) incentive pay, which is outside of our model. Following arguments in [Taylor \(2013\)](#), we assume that the wage level set each year induces the manager to deploy their human capital at the firm and produce, and any negotiated future incentive today is incorporated into the current piece-rate.

We identify each labor market event for managers in our sample. Executives can be internally promoted, externally promoted, or move horizontally to be an executive at another firm. CEOs can move horizontally to other CEO positions. All managers can enter unattachment or retire. Unattached managers can enter executive or CEO employment.

Table 1 displays summary statistics (Panel A) and labor market transitions statistics (Panel B) for our estimation sample. Wages are deflated to 2010 dollars, and statistics are displayed conditional on employment state (executive, CEO, or unattached). The median CEO in our sample receives about 3 million dollars, with the median executive receiving about 959 thousand dollars. Wage growth is on average about 10.4% (13.4%) per year for CEOs (executives).

External CEO hires are rare relative to internal promotions (575 total external CEO hires relative to 4,568 promotions). These differences could reflect both the importance of firm-specific human capital and different search probabilities for internal and external candidates. An important aspect of our identification and estimation will be separating these two sources. There are a relatively high number of executive-to-executive transitions. As firm-specific human capital is largely identified by changes in managerial wages conditional on labor market experience, and position and executive fixed effects, these transitions prove useful.

¹⁰The data does not allow us to perfectly separate unattachment and retirement. Indeed, any manager who initially enters unattachment (and is still looking for managerial positions), but ultimately never shows up again in Execucomp will be labeled as retirement. As such, it is likely that our separation rate η is biased downward, and our retirement rate μ is biased upwards. This has implications for the stock of managers that firms consider for unfilled positions. However, this does not impact within- and across-job bargaining, mobility and wages.

3. Model

3.1. Environment

The labor market consists of firms and managers. Time is continuous. Firms are modeled as a selection of managerial positions that are either filled or looking to be filled. The variables t and $\tau \leq t$ respectively denote a manager's labor market experience and tenure at their current firm. Managers may be employed in CEO or non-CEO positions, where we refer to the latter type of manager as an "executive." Employed CEOs and executives are compensated in proportion to their contribution to firm output: a manager's total compensation is $W(p, t, \tau)$, where p is a position-specific match productivity parameter. A manager acquires a new p any time they switch positions, and τ is reset to 0 any time a manager leaves their firm. Separation occurs at rate η , in which case the manager enters "unattachment," and retirement occurs at rate μ .

Production and human capital. A manager's total human capital (in logs) is given by:

$$h(t, \tau) = a + g(t) + k(\tau) \quad (1)$$

The parameter $a \sim N(0, \sigma_a^2)$ is an inherent manager-specific skill parameter reflecting permanent differences in individual ability. The functions $g(t)$ and $k(\tau)$ denote general and firm-specific human capital accumulated through work experience. General human capital $g(t)$ can be transferred across firms while position-specific human capital $k(\tau)$ cannot. In other words, if a manager enters unattachment or switches firms, $k(\tau)$ is reset to $k(0) = 0$.

Each manager-position match generates log output:

$$y(p, t, \tau) = p + h(t, \tau), \quad (2)$$

where $p \in [p_{min}, p_{max}]$, $p \sim F(\cdot)$ reflects position-specific match productivity. In our setup, firm-specific human capital increases the match-specific productivity $p + k(\tau)$ over the executive's tenure (Gao et al., 2021). As will be discussed later in this section, this increasing match produc-

tivity will influence how firms bargain over executives at different stages of tenure.

When a manager is not employed as an executive or CEO and is instead unattached, their tenure τ is fixed at zero and firm-specific human capital does not accumulate. We allow experience to accumulate while in the unattachment state. The nature of the data on executive wages and mobility necessarily limits us to analyzing publicly-listed firms. Given the nature of the executive labor market, it is likely that unattached executives are accumulating experience, for example by managing a private firm. Upon entering (observable) managerial employment, a manager's tenure and firm-specific human capital immediately begin to accumulate.

Retirement, unattachment and job search. All managers engage in on-the-job search and may make contact with a potential employer about a new position at any time. Upon contact, match-specific type p is drawn from $F(\cdot)$, with support $[p_{min}, p_{max}]$.

Differing levels of seniority impact the types of offers managers field. For executives, three possible offers may be received: internal promotion to CEO, external promotion to CEO, or a horizontal move to another executive position. Internal promotion opportunities arrive at rate λ_0 . If a manager is considered for internal promotion, we assume that they draw a new position productivity p .¹¹ The key difference between internal and external mobility is that firm-specific human capital is preserved and continues to accumulate following an internal promotion.

External promotions arrive at rate λ_1 . If accepted, the executive switches firms, takes the new CEO position of type p and their firm-specific capital resets to zero. Differences in the arrival rates of internal and external promotions (λ_0, λ_1) may reflect, for example, preferences (for or against) hiring CEOs within the firm or differences in firms' cost of internal and external search for executive talent. Lastly, offers for an executive position at a different firm arrive at rate λ_2 . If such an offer arrives and is accepted, firm-specific human capital will again reset to zero.

For a manager employed as CEO, the only types of viable outside offers are for CEO positions

¹¹This assumption captures the idea that for a given manager, their fitness for the CEO position may differ with that of another c-suite position. As explained below, when evaluating a promotion opportunity, the executive and firm will internalize this difference in productivity when they begin to bargain.

at different firms; CEOs cannot be internally or externally demoted.¹² We assume that the arrival rate λ_1 is the same for both executives and CEOs. We will show however that despite this assumption, the rates of executive-to-CEO and CEO-to-CEO transitions will differ in equilibrium. As in the case for executives, if an outside offer is accepted, p is redrawn and $k(\tau)$ resets to zero.

When a manager sees their match dissolved, they enter a period of unattachment. While in the unattachment state, managers may receive offers for executive or CEO positions at rates γ_E and γ_C , respectively. Unattachment is not synonymous with unemployment. When unattached, managers have no pecuniary tie to any publicly-held firm, but may still seek employment in the government or private sector, for example. Explicitly modeling these alternative labor markets is beyond the scope of this paper, and we emphasize that the important point for our purposes is that while unattached, managers accumulate no human capital specific to any publicly-held firm in our sample.

3.2. Bargaining and Wage Contracts

Wages are modeled as piece-rate contracts in which managers receive an endogenous share $R \in (0, 1]$ of their marginal product (Postel-Vinay and Robin, 2002; Cahuc et al., 2006; Bagger et al., 2014). Specifically, the level of compensation $W(p; t, \tau)$ is given by:

$$W(p; t, \tau) = Re^{y(p, t, \tau)}$$

or in log form:

$$w(p, t, \tau) = r + p + h(t, \tau) = r + p + a + k(\tau) + g(t) \quad (3)$$

Piece rates $r = \log(R) \leq 0$ are determined via Nash bargaining where $r = 0$ captures the extreme case in which the manager fully extracts the surplus generated by the position. For a manager

¹²This assumption is motivated by the data. When examining potential demotions (when a manager is the CEO at a firm in year t and a non-CEO executive in year $t + 1$) in our data, we find that a large majority comprise advisory positions. For example, the previous CEO stays on at the firm explicitly as an advisor to the current CEO, or implicitly by taking chairmanship of the board. However, a small number of true demotions do exist in our data. Our model is not intended to study the decision to promote a (possibly interim) CEO and subsequently demote them, so we remove these managerial spells from our estimation sample.

with experience t , tenure τ , and match productivity p under a contract stipulating piece rate r , we denote the discounted value of their position by $V_i(r, h, k, p)$, for $i \in \{E, C\}$.¹³

Over their careers, managers leverage competing offers to increase the values of their current positions (through renegotiation) or transition into more valuable positions. When a job offer arrives, a bargaining game is initiated between the manager and firm(s) bidding for their services.

The bargaining protocol can be seen as an extension of that found in Cahuc et al. (2006), which we provide microfoundations for in Appendix A. First, we relax the assumption of perfect transferability of human capital, and show that this has key implications for wage trajectories and employment transition probabilities in equilibrium. Second, our bargaining game is complicated by the fact that over their careers, managers may receive offers for both executive and CEO positions. This introduces a set of trade-offs across position types which managers must consider as they climb the job ladder, and allows us to make theoretical predictions about the dependence of CEO pay on prior employment histories. We begin this discussion by next describing the bargaining protocol determining the evolution of the piece rate r .

3.2.1. Managerial Bargaining

Our model entails three different forms of bargaining. Non-CEO executives can be promoted to CEO internally or receive external offers to be CEO. Both executives and CEOs can receive horizontal offers. We detail each of these bargaining protocols in turn.

Internal promotions. Consider an executive with state (r, h, k, p) and suppose their firm approaches them offering to promote them to CEO. The executive's position productivity for the CEO role p' is drawn from the distribution $F(\cdot)$; p' can be greater or smaller than p , reflecting that managers are more or less productive within different roles at the same firm.

The new match productivity p' may be low enough that the manager and firm pass on the promotion and the executive stays in their current role. Otherwise, the firm and executive bargain

¹³Experience t is kept implicit in the state to reduce notational clutter. Additionally, including $k(\tau)$ in the state is equivalent to including τ , as $k(\tau)$ is sufficient for τ .

and the outcome is a piece rate r' which satisfies the following condition:

$$\begin{aligned} V_C(r', h, k, p') &= V_E(r, h, k, p) + \beta_0 [V_C(0, h, k, p') - V_E(r, h, k, p)] \\ &= \beta_0 V_C(0, h, k, p') + (1 - \beta_0) V_E(r, h, k, p) \end{aligned} \quad (4)$$

The value to the manager of accepting the internal promotion is the value of the current match plus a share β_0 of the additional rents arising out of the new position. The parameter β_0 , to be estimated, measures managers' internal bargaining power.

Note that the sharing rule implies that it is in the executive's interest to accept the promotion to CEO if and only if $V_C(0, h, k, p') > V_E(r, h, k, p)$. It will be useful to define $\bar{\theta}(r, k, p)$ as the critical level of match productivity such that the manager would accept an internal promotion:

$$V_C(0, h, k, \bar{\theta}(r, k, p)) = V_E(r, h, k, p) \quad (5)$$

Given an executive with state (r, h, k, p) , if $p' \geq \bar{\theta}(r, k, p)$, the executive accepts the CEO role under the wage contract defining (4). Otherwise, if $p' < \bar{\theta}(r, k, p)$, the executive and firm halt their negotiations and the executive remains in their current position. We assume that $\bar{\theta}(p_{min}) = p_{min}$, in which case the following result holds:

Theorem 1. $\bar{\theta}(r, k, p) \leq p$. Executives would be willing to accept a promotion to CEO at their current firm, even if that promotion entails a loss in match productivity.

Proof. See Appendix A. ■

This result follows from the structure of the managerial job ladder. CEOs, having already made it to the top, have less opportunity for vertical mobility. So, in equilibrium, CEO employment is more valuable than executive employment, all else equal, because expected tenure is endogenously longer for CEOs. Executives thus benefit from accepting a promotion to CEO even if this means moving into a position in which they are less productive.

External promotions. Next, consider an executive with state (r, h, k, p) who is approached by a poaching firm with an offer to serve as CEO. Again, a match-productivity p' is drawn and the manager, their current firm, and the poaching firm initiate a bargaining game. Relative to the case of internal offers, managers find themselves in more favorable bargaining positions upon the arrival of external offers, as they can pit their current and poaching firm against one another (Cahuc et al., 2006). Such competition over their services has a positive effect on the manager's threat point. We show in Appendix A that if $p' > \bar{\theta}(r, k, p) + k(\tau)$, the manager accepts the external promotion under a piece rate r' satisfying:

$$V_C(r', h, 0, p') = \beta_1 V_C(0, h, 0, p') + (1 - \beta_1) V_E(0, h, k, p) \quad (6)$$

where β_1 measures managers' external bargaining power, which we do not restrict to equal β_0 . Notice that given the same state (r, h, k, p) , the minimum level of productivity required to accept an external CEO position is larger than that of an internal CEO position. This arises by virtue of the fact that firm-specific capital causes the match-specific productivity to increase over tenure: $k(\tau)$ is not transferable across firms and upon accepting an offer with a new firm, k resets to zero. The match productivity associated with the new position (p') must be large enough to compensate for this lost firm-specific human capital. If $p' < \bar{\theta}(r, k, p) + k_-$, where k_- denotes the executive's firm-specific human capital at their current firm just before the offer arrives, the poacher cannot successfully hire the executive. Despite this, the manager may still use this poaching offer as leverage and renegotiate their pay contract with the incumbent firm. Namely, if $p' < \bar{\theta}(r, k, p) + k$, the incumbent firm retains the manager by adjusting their piece rate according to:

$$V_E(r', h, k, p) = \beta_1 V_E(0, h, k, p) + (1 - \beta_1) V_C(0, h, 0, p') \quad (7)$$

Though a given offer may not be attractive enough for the manager to accept in equilibrium, the *threat* of acceptance triggers a bidding war between the poaching and incumbent firms, ultimately increasing the manager's share of surplus. Lastly, an outside offer for a CEO appointment may

be so unattractive as to not trigger a change in the current piece rate r . We denote $\underline{\theta}_O$ as the minimum level of productivity which triggers a piece rate renegotiation:

$$V_E(r, h, k, p) = \beta_1 V_E(0, h, k, p) + (1 - \beta_1) V_C(0, h, 0, \underline{\theta}_O) \quad (8)$$

Horizontal Moves. Finally, managers can receive offers to serve in an equivalent role at another firm, which we refer to as a “horizontal” move. The bargaining protocol works largely the same as in the previous case, though somewhat simpler, as there is no trade-off between position types. Consider a manager in position type $i \in \{E, C\}$ with state (r, h, k, p) who is approached by a poaching firm with an offer to serve in an equivalent role. A new match-productivity p' is drawn the manager, their current firm, and the poaching firm begin bargaining. If $p' > p + k$, the manager accepts the executive role with the poaching firm under r' satisfying:

$$V_i(r', h, 0, p') = \beta_1 V_i(0, h, 0, p') + (1 - \beta_1) V_i(0, h, k, p) \quad \text{for } i \in \{E, C\} \quad (9)$$

Conversely, if $p' \leq p + k$, the incumbent firm retains the manager by offering a contract satisfying:

$$V_i(r', h, k, p) = \beta_1 V_i(0, h, k, p) + (1 - \beta_1) V_i(0, h, 0, p') \quad \text{for } i \in \{E, C\} \quad (10)$$

As before, competing offers may be so unattractive as to not induce any renegotiations with the manager’s current employer. Define the minimum level of productivity which triggers a renegotiation as:

$$V_i(r, h, \tau, p) = \beta_1 V_i(0, h, \tau, p) + (1 - \beta_1) V_i(0, h, 0, \underline{\theta}_i) \quad \text{for } i \in \{E, C\} \quad (11)$$

3.2.2. Unattached Managers

When unattached, managers can receive an executive or CEO job offer (with rates γ_E and γ_C). We assume throughout the paper that the value of being unattached is equivalent to employment in

the least-productive executive position, $V_U(h) = V_E(0, h, 0, p_{min})$. As in [Bagger et al. \(2014\)](#), this convenient assumption means that the unattached manager accepts any job offer.

3.3. Value Functions

We assume that managers have logarithmic flow utility and that there is no transfer of wealth across time. All parties discount the future at rate ρ . Let $S(\cdot) = 1 - F(\cdot)$ be the survivor function for the distribution of position types. Given the threshold position $\underline{\theta}$ which leads to no contract revision, $S(\underline{\theta})$ represents the fraction of positions for which the manager discards the job offer.

CEOs have the value function:

$$\begin{aligned}
 (\rho + \mu + \eta + \lambda_1 S(\underline{\theta}_C)) V_C(r, h, k, p) = & w + \eta V_U(h) + \\
 & \lambda_1 \int_{p+k}^{p_{max}} \mathbb{E} \left\{ (1 - \beta_1) V_C(0, h, k, p) + \beta_1 V_C(0, h', 0, x) \right\} dF(x) + \\
 & \lambda_1 \int_{\underline{\theta}_C}^{p+k} \mathbb{E} \left\{ (1 - \beta_1) V_C(0, h', 0, x) + \beta_1 V_C(0, h, k, p) \right\} dF(x)
 \end{aligned} \tag{12}$$

The net present value of holding a CEO position is the sum of flow compensation and expectations over future employment transitions. Note that the integral terms in the value function above reflect the structure of the CEO bargaining process. If a poaching offer with $p' > p + k$ arrives, the CEO transitions to the poaching firm and receives a piece rate satisfying Equation (9). If a poaching offer with $p' \in \{\underline{\theta}_C, p + k\}$ arrives, the CEO remains with their current firm at a renegotiated rate defined by Equation (10). Lastly, if $p' < \underline{\theta}_C$, the poaching offer is discarded and nothing happens. The net present value of holding an executive position, though slightly more cumbersome, is similar in structure:

$$\left(\rho + \mu + \eta + \sum_{s=0}^2 \lambda_s S(\underline{\theta}_s) \right) V_E(r, h, k, p) = w + \eta V_U(h) +$$

$$\begin{aligned}
& \lambda_0 \int_{\bar{\theta}}^{p_{max}} \mathbb{E} \left\{ (1 - \beta_0) V_E(0, h, k, p) + \beta_0 V_C(0, h, k, x) \right\} dF(x) + \\
& \lambda_1 \int_{\bar{\theta}+k}^{p_{max}} \mathbb{E} \left\{ (1 - \beta_1) V_E(0, h', k', p) + \beta_1 V_C(0, h', 0, x) \right\} dF(x) + \\
& \lambda_1 \int_{\underline{\theta}_O}^{\bar{\theta}+k} \mathbb{E} \left\{ (1 - \beta_1) V_C(0, h', 0, x) + \beta_1 V_E(0, h', k', p) \right\} dF(x) + \\
& \lambda_2 \int_{p+k}^{p_{max}} \mathbb{E} \left\{ (1 - \beta_1) V_E(0, h', k', p) + \beta_1 V_E(0, h', 0, x) \right\} dF(x) + \\
& \lambda_2 \int_{\underline{\theta}_2}^{p+k} \mathbb{E} \left\{ (1 - \beta_1) V_E(0, h', 0, x) + \beta_1 V_E(0, h', k', p) \right\} dF(x)
\end{aligned} \tag{13}$$

3.4. Equilibrium Contracts

We next summarize the equilibrium piece rate contracts, paying special attention to how the transferability of human capital across firms impacts managerial wage growth and mobility. We begin by analyzing how the executive contract differs across the mobility events over an executive's career. That is, how does the contract look when an executive has accepted a new position, and what forces within the model drive differences in executive surplus capture across different career events.

3.4.1. The CEO contract

We define the CEO's piece rate by the function $r_C(p, k, \underline{\theta}_C)$, which depends on the CEO's current position productivity, their level of firm-specific human capital, and the threshold match quality $\underline{\theta}_C$ which tracks the last match quality from which the CEO captured the full surplus in the bargaining game. Contracts are restructured only in the event of a competing offer. Depending on the match productivity associated with the competing offer, three possibilities may be realized. First, if $p' > p + k$, the manager accepts the offer and vacates their current position. Second, if $p' \in [\underline{\theta}_C, p + k]$, the manager remains in their current position contingent on renegotiating the terms of their contract. Finally, if $p' < \underline{\theta}_C$, the CEO discards the offer and the contractual terms remain unchanged. We begin by analyzing transitions of employment, so that the potential new

position p' triggers a job switch.

Mobility. We adopt the following terminology when describing different types of CEO transitions: *horizontal* moves refer to across-firm CEO to CEO transitions, *diagonal* moves refer to across-firm executive to CEO transitions, and *vertical* moves refer to within-firm executive to CEO transitions. We begin by analyzing the case of horizontal CEO transitions. Consider a CEO employed in a p position who is poached into a p' position. Denote by k_- the CEO's level of firm-specific human capital just before leaving their previous firm. The initial piece rate in their new position is given by:

$$r_C^{hor}(p', 0, p + k_-) = -(1 - \beta_1) \int_{p+k_-}^{p'} q(x) dx \quad (14)$$

where

$$q(x) = \frac{\rho + \mu + \eta + \lambda_1 \bar{F}(x)}{\rho + \mu + \eta + \lambda_1 \beta_1 \bar{F}(x)}$$

This expression resembles the piece rate from [Bagger et al. \(2014\)](#), but importantly differs because of its dependence on k_- . With imperfect transferability of human capital, the CEO's firm-specific capital in their previous position has no direct impact on output in their new position yet still directly impacts the structure of compensation. Namely, the new piece rate increases with respect to the CEO's tenure in their previous position. Thus, long-tenured CEOs with high levels of firm-specific proficiency are more expensive to poach than their newly-tenure counterparts.

Consider next an executive in a p position who is externally promoted into a p' -productivity CEO position. Their initial piece rate is given by:

$$r_C^{diag}(p', 0, p + k_-) = r_C^{hor}(p', 0, p' + k_-) - (1 - \beta_1) \int_{\bar{\theta}(p)+k_-}^{p+k_-} q(x) dx \quad (15)$$

Theorem 1 implies that the second term in (15) is negative. Hence, when considering externally-hired CEOs, those appointed from executive positions are initially paid less than those appointed

from CEO positions (all else equal). This pay gap reflects differences in outside options between these two types of manager at the time of the wage negotiation: because executive positions are less valuable sources of employment than CEO positions, it is cheaper for firms to successfully poach an executive than a CEO.

Finally, consider an executive in a p position who is internally promoted into a p' CEO position. The initial piece rate is given by:

$$\begin{aligned} r_C^{vert}(p', 0, p' + k_-) &= r_C^{diag}(p', 0, p' + k_-) - (1 - \beta_1) \int_{p'}^{p' + k_-} q(x) dx \\ &\quad + (\beta_0 - \beta_1) \int_{\bar{\theta}(p) + k_-}^{p'} \frac{\rho + \mu + \eta}{\rho + \mu + \eta + \lambda_1 \beta_1 \bar{F}(x)} dx \end{aligned} \quad (16)$$

When comparing vertical and diagonal promotions to CEO, the above condition highlights the mechanisms yielding disparities in their initial pay contracts. First, as vertically-promoted CEOs retain their accumulated firm-specific human capital upon accepting the promotion, firms do not need to offer as much surplus in order to win their services (reflected in the second term in the equation above). Additionally, the two types of CEOs command different levels of bargaining power, so receive different shares of surplus when bargaining. If $\beta_0 \leq \beta_1$, the internal CEOs are. The relative level of $r_C^{vert}(p', 0, p' + k_-)$ and $r_C^{diag}(p', 0, p' + k_-)$ in general depends on the relative levels of bargaining power between the two CEO types, and as such must be resolved empirically.

Renegotiation. Next, we analyze the effect of outside CEO offers with match productivity $p' \in [\underline{\theta}_C(p, k), p + k]$, in which case managers are retained by contracts are renegotiated. We refer to the set $[\underline{\theta}_C(p, k), p + k]$ as the *renegotiation region*, the size of which is dependent on CEO type. This represents the set of competing positions which, while not lucrative enough for the CEO to accept, would provide the CEO with some leverage over their current employer in the event of an offer arriving. Upon receiving an outside offer with productivity p' in this range,

the renegotiated piece rate will be:

$$r_C(p, k, p') = -(1 - \beta_1) \int_{p'}^{p+k} q(x) dx \quad (17)$$

Of course, it is in the CEO's interest to discard offers which would yield negative piece rate revisions following from renegotiations with their employer. In order to trigger a renegotiation, a competing offer must have productivity exceeding $\underline{\theta}_C(p, k)$, as defined in Equation (11). $\underline{\theta}_C(p, k)$ defines the lower boundary of the CEO's renegotiation set, whose dynamics are summarized in the following theorem:

Theorem 2. Define $\Omega_r^m(p, k)$ as the mass of the renegotiation region for each $m \in \{hor, diag, vert\}$.

For all m :

$$\frac{\partial \Omega_r^m}{\partial k}(p, k) > 0 \quad (18)$$

That is, the renegotiation region grows in size as firm-specific human capital accumulates. Additionally, given a value of p , we have for all k :

$$\Omega_r^{vert}(p, k) > \Omega_r^{diag}(p, k) > \Omega_r^{hor}(p, k) \quad (19)$$

Proof. See Appendix A. ■

The above result has important empirical implications. First, as k grows with CEO tenure, the probability of an employment transition gradually declines. CEOs experience a growing degree of “job lock;” because switching positions entails sacrificing their firm-specific human capital, CEOs grow increasingly willing to remain in their current position as opposed to seeking employment elsewhere. However, the probability of contractual renegotiations grows with tenure, as the mass of the renegotiation region is increasing. By virtue of firm-specific human capital accumulation, CEOs grow increasingly valuable to their employers with tenure. Their employers

are thus increasingly willing to concede favorable renegotiations to the CEO's terms of contract as not to lose them to a competing firm.

Moreover, the likelihood of favorable renegotiation varies by CEO type. As vertically-hired CEOs retain their firm-specific human capital upon promotion, they are the most valuable of the three types for a given level of p . Thus, these types hold an especially high amount of leverage over their employers, and should see relatively higher rates of renegotiation. Horizontally-hired CEOs, who bring no firm-specific human capital to the table and are relatively expensive to hire, are the least likely to see favorable renegotiations conceded from their employer. Diagonally-hired CEOs, who also begin employment with no firm-specific human capital but are cheaper to hire than horizontal CEOs, lie somewhere in between the two extremes.

3.4.2. The executive contract

Our model also admits a definition for the executive piece rate, which tracks how vertical (internal CEO promotion), diagonal (external CEO promotion) and horizontal (external executive) opportunities impact the executive contract throughout their careers and tenures.

Analyzing executive (not just CEO) contracting is important theoretically, because it allows us to model directly internal promotions. Further, when we undertake the estimation, executive-to-executive transitions will be important for separately identifying general and firm-specific human capital.

Despite this, analyzing the executive contract in detail is less important than the CEO contract from a prediction perspective. Nevertheless, we show in Appendix A that the executive piece rate, upon switching to a p' position from a p position with k_- firm-specific human capital can be written as:

$$r_E(p', 0, p + k_-) = -(1 - \beta_1) \left\{ \int_{p+k_-}^{p'} q_E(x) dx - \lambda_0 \beta_0 \int_{\tilde{\theta}(p)}^{\tilde{\theta}(p')} \phi_C(x) S(x) dx + \lambda_1 (1 - \beta_1) \int_{\tilde{\theta}(p)+k_-}^{\tilde{\theta}(p')} \phi_C(x) S(x) dx \right\} \quad (20)$$

where

$$q_E(x) = \frac{\rho + \mu + \eta + \lambda_0 \beta_0 S(\bar{\theta}(x)) + \lambda_1 \beta_1 S(x) + \lambda_2 S(x)}{\rho + \mu + \eta + \lambda_0 \beta_0 S(\bar{\theta}(x)) + \lambda_1 \beta_1 S(x) + \lambda_2 \beta_1 S(x)}$$

$$\phi_C(x) = \frac{1}{\rho + \mu + \eta + \lambda_1 \beta_1 S(x)} = \frac{\partial V_C}{\partial x}(0, h, 0, x)$$

The first line is the executive-equivalent to the CEO piece rate, and reflects the impact of horizontal mobility on the executive contract. Where the executive contract differs is in how *promotion opportunities*, both internal and external impact the executive's contract.¹⁴

3.4.3. The managerial wage process

The analysis above on contracting in our model gives the empirical wage processes for managers (for $j \in \{E, C\}$):

$$w_j(p, t, \tau) = \alpha + \varepsilon(t) + g(t) + k(\tau) + p + r_j(p, k(\tau), \underline{\theta}_j) \quad (21)$$

We can use (21) to write the model's wage growth process over the manager's career, which allows us to decompose wage growth into its various pieces. First, consider managers (both executives and CEOs, $j \in \{E, C\}$) who do not enter a state of unattachment and denote p' as the (potential) draw for a new position from search. The variable $\underline{\theta}_j'$ tracks the possible new threshold threshold position productivity for which the manager would capture the full surplus, and the variable $\underline{\theta}_j$ (implicitly a function of p and τ) tracks the current threshold productivity. Conditional on the manager remaining in their current state of employment (executive or CEO employment)¹⁵, the managerial wage growth process is

$$dw_j(p, t, \tau) = \underbrace{dg(t)}_{\text{General human capital accumulation}} +$$

¹⁴In Appendix A, we derive the mass of the renegotiation region for the executive and compare it to that of the CEO. We note that it leads to similar predictions as the CEO equivalent.

¹⁵Eq. (22) is not intended to make statements about transitions across employment states. It is not relevant when a manager switches to unattachment, nor does it make predictions about executive promotions to CEO.

$$\begin{aligned}
& \underbrace{dk(\tau) dN_t(1 - \lambda_j F(p + k(\tau)))}_{\text{Firm-specific human capital accumulation}} + \\
& \underbrace{\left(r_j(p, k(\tau), \underline{\theta}_j') - r_j(p, k(\tau), \underline{\theta}_j) \right) dN_t(\lambda_j(F(p + k(\tau)) - F(\underline{\theta}_j)))}_{\text{Contract renegotiation}} + \\
& \underbrace{\left(p' - (p + k(\tau)) + r_j(p, 0, p + k(\tau)) - r_j(p, k(\tau), \underline{\theta}_j) \right) dN_t(\lambda_j S(p + k(\tau)))}_{\text{Job-hopping}}
\end{aligned} \tag{22}$$

where $dN_t(x)$ is a Poisson process with arrival rate x .¹⁶

Eq. 22 is a key equation of the paper. The first line tells us the impact of general human capital accumulation on wage growth, the portion of human capital that stays with the manager across their career. The second line tracks firm-specific human capital accumulation (note that firm-specific human capital will accumulate in all states when the manager does not switch firms). The third line tracks the impact of job-hopping: managers taking up other positions over their career (where we incorporate the loss of firm-specific capital in this event). The last line tracks the impact of on-the-job search on managerial wages over their career — job offers that lead to contract renegotiation, but ultimately do not induce the CEO to switch firms.

The empirical value of this equation is apparent. The relative importance of the components of human capital is largely unobservable in the data. Further, any impact of search on contract renegotiation is also unobservable.

3.5. Model predictions

Our model makes several predictions about the impact of firm-specific human capital executives wages and mobility. While the estimation results will ultimately reveal the importance of firm-

¹⁶Conditional on remaining employed, two events may lead to a job-hob or a contract renegotiation. These events arrive with intensity $\lambda_j[F(p + k(\tau)) - F(\underline{\theta}_C)]$ and $\lambda_j[S(p + k(\tau))]$, respectively. Note that job-hopping discretely impacts a CEO's stock of firm-specific capital, so with complementary intensity (relative to job-hopping), the CEO continues to accumulate human capital.

specific capital, illustrating these predictions is useful for summarizing the theoretical results from the model and guiding the estimation.

Prediction 1. *Firm-specific human capital makes it less likely that a manager leaves their current firm, i.e. mobility across firms decreases.*

This prediction immediately follows from the upper bounds of the renegotiation region for external mobility events; for both CEOs and executives, this upper threshold is increasing in k . The evidence in [Cziraki and Jenter \(2024\)](#) suggests that firm-specific human capital plays a larger role in the market for CEOs than previously thought. Our model delivers the prediction that it plays an important role in determining the cross-firm mobility of executives.

Prediction 2. *Firm-specific human capital makes it more likely that a manager receives a job offer that leads to a renegotiation at their current firm, i.e. mobility within-firm increases.*

This prediction follows directly from Theorem 2, which shows that the renegotiation region grows as firm-specific human capital accumulates. A direct implication of this prediction, given (22), is that contract renegotiation plays a more important role (at least on the extensive margin) in wage growth than job-hopping as firm-specific capital accumulates. While the market for managers implies that job-switching is relatively rarer given firm-specific skill, it *does* imply that on-the-job search and market forces can induce positive changes in wages via managers bargaining with their current firm, and firm-specific human capital plays an important role in this process.

Prediction 3. *The impact of on-the-job search for CEOs is path-dependent, and depends on whether executives are promoted internally, promoted externally, or horizontally poached.*

A subtle follow-on implication of Theorem 2 is that on-the-job search has different implications for depending on how the CEO was hired. In particular, the renegotiation region (given productivity p) is always the largest for vertically-promoted CEOs. This is exactly because firm-specific skill survives the internal promotion process. While horizontally poached CEO-to-CEO

transitions are generally more expensive (firms must compensate the CEO for non-transferable firm-specific skill), they also are the least likely to see renegotiations arising from firm-specific capital.

Prediction 4. *Holding all else equal, including the position productivity p , poached CEOs garner the most favorable contracts. The difference in contracts for diagonally- and vertically-hired CEOs will depend on the relative magnitude of internal and external managerial bargaining power.*

This prediction concerns the search-driven bargaining power of newly-hired CEOs. Poached CEOs require the largest piece rates to tempt them from their current positions. In order to compensate CEOs for the value lost from giving up their firm-specific skill, the contracts must be larger. Our estimation will reveal the the contracting differences between diagonally-poached and vertically-promoted CEOs.

4. Estimation

We estimate the model via indirect inference (Smith, 2016).¹⁷ Appendix B details the estimation algorithm and the majority of this section is devoted to our identification strategy.

4.1. Model solution and simulation

The set of managerial contracts described in Section 3.4 provide a full characterization of the data generating process. For any manager (executive or CEO), the equilibrium sharing rules and the wage equation (21) describe how executive wages evolve across experience, tenure, and the possibly mobility events in our model.

To simulate the model, we start with an initial cross-section of I managers at time 0, and assign them an ability $a \sim N(0, \sigma_a^2)$, and then subsequently assign them to executive and CEO positions

¹⁷Indirect inference is a simulated method of moments estimator which involves fitting an auxiliary model that incorporates parameters from (potentially misspecified) reduced-form econometric models. This approach entails comparing the parameters of the auxiliary model fitted on both observed and simulated data to find the vector of structural parameters that minimizes the distance between them. Indirect inferences generates consistent estimates of the economic model by mapping the parameters of the structural (economic) model to the parameters of the auxiliary model.

based on the rates γ_E and γ_C as if they come from unattachment. For each employed manager, we draw an initial p productivity and assume that the lower bound for renegotiation is p_{min} upon birth. As in [Bagger et al. \(2014\)](#), we discretize the productivity grid into P points and assign each point a weight according to the density function of the underlying Weibull distribution.

From there, labor market events (in/out of unattachment, mobility) arrive at the relevant rates for each manager at any given time. For each job offer, a position productivity p' is drawn and the piece rate is updated given the relevant labor market outcome (job-switch, renegotiation, no change). Retired managers are replenished at the rate μ to keep the stock of managers (in CEO and executive positions) steady across time. Depending on the job offer (or if none arrives), the lower bounds for renegotiation and piece rates are updated. Human capital accumulates over time and contributes to changes in wages.

We simulate the model for 80 years, and use the first 50 years as “burn-in” to allow the economy to approach steady-state.¹⁸ Once a worker reaches 45 years of experience, we force retirement. As such, we are left with a panel data with 30 years of data and a cross-section of managers, which allows for accurate comparison with the data moments.

4.2. Identification Strategy

We show there is a tight relation between the reduced-form outcomes of the auxiliary model and structural parameters, which is key to the success of the indirect inference approach. We have three sets of “moments” which target different sets of structural parameters, and our identification argument is inspired by key papers in the structural search and human capital literature (e.g., [Cahuc et al., 2006](#); [Bagger et al., 2014](#)); crucial to its success is separating the impacts of search frictions, imperfect labor market competition, and firm-specific human capital accumulation in determining compensation and mobility over the executive’s career.

¹⁸Structural papers in the search literature often derive the steady-state distributions and use that to simulate economies. While an analytical description of the steady-state of our model may exist, it requires full derivations of, for example, the conditional distribution of position types p and job types E, C given both experience and tenure, the conditional distribution of tenure given experience, and a conditional distribution of piece rates (renegotiation thresholds) given position-type, job-type and tenure. As our ultimate goal is empirical, we prefer to let the simulation arrive to steady-state.

4.2.1. Executive Labor Market Mobility

The model entails three types of (un)employment: unattachment, executive employment and CEO employment. We allow all conditional transition probabilities (from employment to unattachment) to vary by tenure. In particular, we condition on managers in the first five years of tenure at their current firm (or later) when calculating the empirical moments that identify labor market mobility, which is empirically motivated.

Managers retire/die at rate μ_τ for any employment type. In the data, if a manager transitions out of their current position and does not appear in the data again, we assume retirement. The parameters μ_τ is identified by the the empirical probabilities that an employed manager (executive or CEO) enters retirement (conditional on tenure). The parameters η_τ is identified by the same for employed managers entering unattachment. We replenish the economy with new unattached managers (experience and tenure of 0) to match the unconditional retirement rate μ

Unattached managers can become an executive, a CEO or stay unattached. The rates γ_E and γ_C capture the likelihood of these events. These are identified by the empirical probabilities of an unattached manager entering executive or CEO employment from unattachment.

Executives (i.e., non-CEOs) can experience three mobility events at any given point (excluding entering unattachment or retirement). They can be promoted to CEO at their current firm ($\lambda_0(\tau)$), become CEO at another firm ($\lambda_1(\tau)$), move horizontally ($\lambda_2(\tau)$). These are identified by the early- and later-tenure empirical probabilities of these events happening to executives. Similarly for CEOs, they can only experience one mobility event: $\lambda_1(\tau)$. Thus, the parameter $\lambda_1(\tau)$ is identified by both the executive and CEO probability of leaving their firm to be CEO at another.

4.2.2. Mincer Wage Regressions

For all employed managers, we estimate a wage equation in the style of [Mincer \(1974\)](#). For firm i , position j , executive s and year t ,

$$w_{ijst} = \sum_{q=1}^Q (a_q \tau_{ist}^q + b_q t_{st}^q) + \delta_{prom} \times \text{Prom}_{ijst} + \delta_{move} \text{Move}_{ijst} + \gamma_{ij} + \phi_s + u_{ijst} \quad (23)$$

Conditional on the position and manager, variation in wages across differing levels of experience in tenure are useful for identifying the $g(t)$ and $k(\tau)$.

The firm $\times$ position (CEO vs. non-CEO) fixed effect γ_{ij} helps identify the position productivity distributions. As in [Bagger et al. \(2014\)](#), we assume a Weibull distribution for productivity in the model, which requires a location p_{min} , a scale and a shape parameter (s_{j1} and s_{j2}), where $F_j(p) = 1 - \exp(-[s_{j1}(p - p_{min})^{s_{j2}}])$.¹⁹ We winsorize γ_{ij} at the 0.05 percentile and include the first four moments of the distribution, by position. The volatilities of the manager-specific fixed effect and the residual help pin down variation in managerial ability and the human capital idiosyncratic shock. To help pin down the AR1 coefficient in the human capital shock, we include as a moment the autocorrelation in u_{ijt} for executives with at least eight years of consecutive experience.

Note that we do not estimate different productivity-type distributions for executive and CEO positions. However, targeting different *conditional* distributions in the data is useful for identification. This is because differences in position-productivity arise endogenously for executives and CEOs solely through the types of job opportunities they receive throughout their career.

The coefficients δ_{prom} and δ_{move} identify the bargaining power parameters β_0 and β_1 . To see this, for executive s , let $N_s(t)$ denote the number of positions s has held up until time t , and similarly let $I_s^0(t)$ denote whether the manager was internally promoted to CEO at time t , and let $N_s^1(t)$ and $N_s^2(t)$ denote the number of external movements at time t (to CEO and to executive, respectively). The executive's current piece rate can be written as:

$$\begin{aligned} r_s(N_s(t)) &= r_s(0) + \sum_{i=1}^{N_s(t)} \overbrace{(r_s(i) - r_s(i-1))}^{\delta_{is}(\beta_0, \beta_1)} \\ &= r_s(0) + I_s^0(t)\delta_{is}(\beta_0) + \sum_{j=1}^{N_s^1(t)} \delta_{js}(\beta_1) + \sum_{k=1}^{N_s^2(t)} \delta_{ks}(\beta_1) \end{aligned}$$

As such, including indicators for whether the CEO was internally promoted (time t or later), and the first period an executive switches jobs identify the bargaining power parameters β_0 and β_1 .

¹⁹The Weibull distribution is appropriate for modeling position-types, as it allows for excess skewness and kurtosis. Given the empirical distribution of executive wages, this is a convenient feature relative to a Normal distribution.

4.2.3. Within- and Across-Job Wage Growth

We also consider how experience and tenure separately impact wage growth of executive careers. We condition only that a manager have two consecutive years of experience to be included in this regression. However, to isolate the impacts of firm-specific human capital on wage growth, we separately study the impacts of experience on tenure, within-job and across mobility events.

Specifically, We define an event-study sample and to focus on when executives change firms. That is, in a 3-year window around the event (so, a maximum of seven years of data per manager, and a minimum of two years experience at either firm in the years before and after the switch), we focus on how changes in wages differ in mobility events. Specifically defining $e_{ijst} = 1$ as an indicator for whether a manager-year observation falls into the event-sample described, we estimate, for firm i , position j , manager s and year t

$$\Delta w_{ijst} = \sum_{e_{ijst} \in \{0,1\}} \sum_{q=2}^Q \left(\omega_{t,q,e} [\Delta t_{st}^q \times e_{ijst}] + \omega_{\tau,q,e} [\Delta \tau_{ist}^q \times e_{ijst}] \right) + \gamma_{ij} + \phi_s + \Delta v_{jst} \quad (24)$$

The slope coefficients in (24) are most useful for identifying $g(\cdot)$ and $k(\cdot)$. We use variation in the change in wages across the tenure that the CEO made the switch to identify how losses in firm-specific capital impact wages. We also include position $\times$ firm and executive fixed effects, which will net out any impact changes in wages arising from differences across positions and executives. We also include the volatility of the residual and the residual autocorrelation to help further separate the human capital shock from the human capital accumulation functions in these two regressions.

It is important to note that (23) and 24 do not have a structural interpretation (Bagger et al., 2014). Indeed, our model would state they are misspecified and suffer from omitted variable bias, as we do not observe the impact of (imperfect) labor market competition on wage growth profiles. Our identification strategy aims to replicate the misspecification and use the structural model to uncover the relevant omitted variables in these regressions.

5. Structural Parameter Estimates

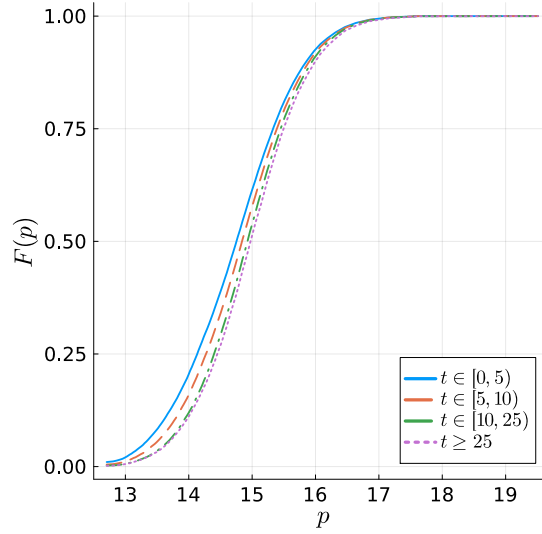
Parameter estimates are reported in Table 2. Regarding transitions into CEO employment, our estimates of λ_0 (.156) and λ_1 (.084) imply that internal CEO offers arrive at roughly twice the rate of external offers. There are several factors which may explain this disparity: the prevalence of non-compete agreements (Shi, 2023), external search costs (Geelen and Hajda, 2024), or non-pecuniary preferences for internal candidates (He and Schroth, 2024; Capron et al., 2024). While this gap is considerable, the gap in internal and external CEO transitions is amplified by limitations in human capital portability.

Our estimates of δ_g (.032) and δ_k (.036) imply that general and firm-specific human capital accumulate at roughly the same rate. The trajectory of total human capital ($h(t, \tau)$), however, is contingent on managers' career path. As an example, consider two managers of equal ability (a), both with 10 years of experience. Total human capital $h(t, \tau)$ will be 18% higher for a manager who stays in the same position relative to one who switched positions in 5th year of their career. For managers, the augmented human capital accumulation stemming from job stability raises the opportunity cost of switching firms and contributes in part to the lack of across-firm mobility observed empirically. This, however, must be weighed against the potential increase in match productivity driven by job shopping.

Our estimates of the distributional parameters s_1 , s_2 , and p_{min} imply that heterogeneity in match productivity (p) is substantial; increasing match productivity from its mean (13.0) by one standard deviation is associated with an 22% increase in total compensation. Further, while we do not impose any direct dependence of p on experience, the equilibrium distributions of match productivity vary systematically with experience. We illustrate this in Figure 1, which plots the cumulative distributions of simulated match productivity separately different levels of managerial experience. In terms of stochastic dominance, match productivity generally increases with experience. Hence, though the market is rigid, managers have some capacity to transition into higher-quality matches over their careers. Endogenous sorting of this nature has important empirical implications. As experience is correlated with unobservable match productivity, naive

Figure 1. Position productivity distribution

This figure displays the estimated position productivity distribution, which is characterized by location p_{min} , shape s_1 and scale s_2 from Table 2.



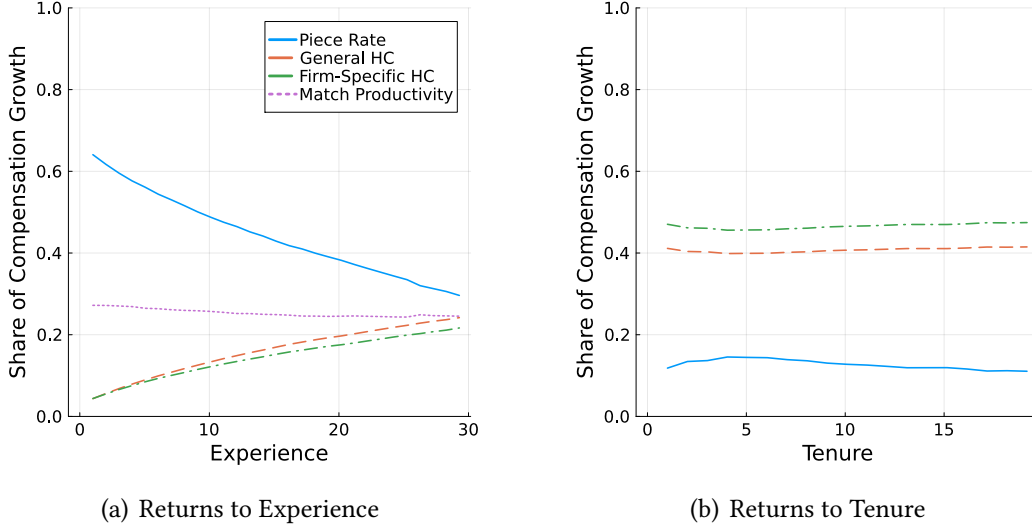
regressions of managerial compensation on experience will yield biased coefficient estimates. To decouple the compensation effects of sorting and human capital accumulation, we decompose CEO compensation growth over experience and tenure in Figure 2.

In Figure 3(a), we first compute average CEO compensation growth for each year of experience. We then compute average growth of each experience-varying component of CEO pay: general human capital, firm-specific human capital, piece rates, and match productivity. Finally, to compute the share of compensation growth attributable to growth in each component, we scale the growth of each component by total compensation growth. The same exercise is done for within-position compensation growth in Figure 3(b), where we omit match productivity as it is fixed within a given position. When explaining the returns to experience, Figure 3(a) suggests that human capital (both general and firm specific) plays a fairly minor role. Increases in the generosity of CEOs' contracts, measured here by the piece rate, explain the largest share of compensation growth over experience, particularly in early stages of careers. In addition, growth in match productivity has a considerable effect on compensation, explaining a stable share of roughly 30% of total compensation growth over the experience profile.

The returns to tenure, illustrated in Figure 3(b), tell quite a different story. Unlike the previous

Figure 2. CEO Compensation Growth Decomposition

This figure accumulated wage growth for CEOs over tenure and experience. We use (22) to decompose accumulated wage growth into changes in the piece rate, general and firm-specific human capital, and position productivity (which does not play a role for tenure).



case, human capital is the main driver of within-position compensation growth, where general human capital has a slight edge. The contribution of piece rate growth is modest and remains stable throughout CEO tenure. The stability of the within-position piece rate contribution suggests that in spite of the job-lock effect, the influence of outside competition on CEO compensation (i.e. renegotiation) is fairly persistent. However, comparing this with the piece rate contribution in Figure 3(a), it becomes apparent that the effect of within-position piece rate renegotiations on compensation growth is far less consequential than across-position piece rate increases realized following job transitions.

6. CEO Bargaining Power and Labor Market Competition

Our estimate of β (.136) implies a modest level of managerial bargaining power. This is markedly lower than previous estimates in the literature.²⁰ Such a low estimate may, at first glance, be difficult to reconcile with the high observed level of managerial pay. Recall however that in our model, managers capture rents not only through their capacity to bargain, but also by leveraging

²⁰For example, Taylor (2013) estimates that CEOs capture about 50% of surplus given upside news about their ability.

across-firm competition for their contracts. Hence, bargaining power and competition work in tandem to increase managerial compensation. By explicitly modeling the influence of outside competition on the setting of managerial pay, we can decouple the effects of competition from pure bargaining power. Put differently, models of managerial bargaining which neglect labor market dynamics, on-the-job search in particular, are susceptible to confounding the effects of competition and bargaining thereby overestimating managerial bargaining power (Cahuc et al., 2006).

This is particularly important in the context of corporate executives. The high observed share of managerial rents is often taken as evidence of managers exercising undue influence in the pay-setting process (Bebchuk and Fried, 2006). Our results suggest that such problems may be overstated, and the high level of rents may instead be the product of well-functioning labor market competition. To illustrate further, we present in this section a decomposition of managers' observed share of rents into two components: those attributable to managers' bargaining power, and those attributable to labor market competition. We then provide an assessment of the bias that results from attempting to estimate CEO bargaining power in the absence of an explicit model of the outside market. In the interest of comparability with previous literature, we restrict attention in this section to those employed in CEO positions.

6.1. Decomposition of CEO Surplus

Our model provides a simple way to separate the compensation effects of bargaining power and competition. Following Cahuc et al. (2006), we derive a “naive” estimate of bargaining power in an alternative version of the model with no on-the-job search, and hence no across-firm competition. Comparing this alternative estimate of bargaining power to our structural estimate from the main model then gives the share of managers' observed surplus which is explained by competitive forces as opposed to pure bargaining power.

This experiment amounts to setting CEOs' on-the-job search parameter λ_1 to zero. In this case, CEOs' outside option is forced to unattachment, which in turn ascribes any CEO surplus

in excess of unattachment value to pure bargaining power.²¹ Consider for example a CEO with match productivity p , tenure τ , and experience t . In this alternative setting with no on-the-job search, the log wage equation must be of the form:

$$w(p, t, \tau) = b(p, \tau, \underline{\theta}_C)(p + k(\tau)) + (1 - b(p, \tau, \underline{\theta}_C))p_{min} + a + g(t), \quad (25)$$

where b measures the CEO's observed share of match surplus. Hence, in the absence of on-the-job search, $\mathbb{E}[b]$ would be our estimate of CEO bargaining power. We can relate this to bargaining power in the main model (β) by setting equal the structural wage equation and the counterfactual wage equation (25):

$$\begin{aligned} b(p, \tau, \underline{\theta}_C) &= \frac{r(p, \tau, \underline{\theta}_C) + p + k(\tau) - p_{min}}{p + k(\tau) - p_{min}} \\ b(p, \tau, \underline{\theta}_C) &= 1 + \frac{r(p, \tau, \underline{\theta}_C)}{p + k(\tau) - p_{min}} \end{aligned} \quad (26)$$

We show Appendix A.1, $b(p, \tau, \underline{\theta}_C) \in [\beta_1, 1]$,²² so the bias that arises when neglecting competition is weakly positive. We can then decompose average total surplus b into a bargaining component and competition component using the simple identity:

$$1 = \underbrace{\mathbb{E}\left[\frac{\beta_1}{b(p, \tau, \underline{\theta}_C)}\right]}_{\text{CEO bargaining power}} + \underbrace{\mathbb{E}\left[\frac{b(p, \tau, \underline{\theta}_C) - \beta_1}{b(p, \tau, \underline{\theta}_C)}\right]}_{\text{labor market competition}} \quad (27)$$

With this decomposition in hand, we estimate $b(p, \tau, \underline{\theta}_C)$ in our simulated sample and report results in Table 4. In the full sample, our naive estimate of CEO bargaining power is roughly .564, which is in line with previously reported estimates in the literature (Taylor, 2013). Comparing this

²¹In our model, unattachment has the same value as an offer from the least productive firm, p_{min} .

²²If $r(p, \tau, \underline{\theta}_C) = 0$, then $b = 1$. The minimum value of $r(p, \tau, \underline{\theta}_C)$ arises when the true λ_1 equals zero, where

$$r(p, \tau, p_{min}) = -(1 - \beta_1)(p + k(\tau) - p_{min})$$

and $b = \beta_1$.

to our estimate of CEO bargaining power β , it is clear that neglecting the impact of competition results in a severe degree of upward bias. Using the formula (27), we find that roughly 75.9% of the observed share of CEO surplus is the product of labor market competition as opposed to pure bargaining power. Thus, despite the across-firm rigidity exhibited in the market for corporate executives, the influence of competition on CEO pay is substantial.

Additionally, we report a number of subsample estimates of $b(p, \tau, \theta_C)$ to illustrate that this bias varies systematically across observables. Panel A of Table 4 reports estimates separately across CEO types, namely whether a given CEO was promoted from within, poached from a CEO position, or poached from an executive position. Panels B and C report estimates across the distributions of experience and tenure, respectively. Starting with the CEO type subsamples, we again see upward bias in the bargaining power estimate in all cases. The magnitude of this bias, however, varies across CEO type. The bias is largest among CEOs who are poached from competing firms, whether from a CEO or executive position. Recall that when attempting to poach a manager, the poaching firm initiates a bidding war with the incumbent. This introduces some degree of competition which the manager can then leverage to negotiate himself a more favorable contract. This is distinct from the internal bargaining regime, in which there is no direct influence of outside competition. Thus, poached managers benefit the most from competition and estimates of bargaining power will be especially biased for these CEO types where competition to be ignored.

Further, we see heterogeneous effects of labor market competition on CEOs' captured share of surplus across both the distribution of experience and tenure. Panel B of Table 4 illustrates that estimates of $b(p, \tau, \theta_C)$ increase monotonically with managerial experience. Put differently, the share of CEO surplus explained by labor market competition increases with experience. This follows from the history-dependent nature of equilibrium piece-rate contracts. The impact of a one-off bidding war has a persistent effect on managers' compensation throughout their careers. Thus, as experience increases and CEOs are subject to a growing number of competitions for their services, these extracted benefits accumulate and comprise an increasing share of their

total captured surplus. Similarly, competition explains an increasing share of CEO surplus as tenure progresses. The rationale for this result is essentially the same as before. As tenure increases, CEOs receive a growing number of outside offers which trigger renegotiations with their incumbent employers. This gradually increases CEOs' share of total surplus, despite their bargaining power remaining constant.

7. Firm-Specific Capital, Mobility, and CEO Rents

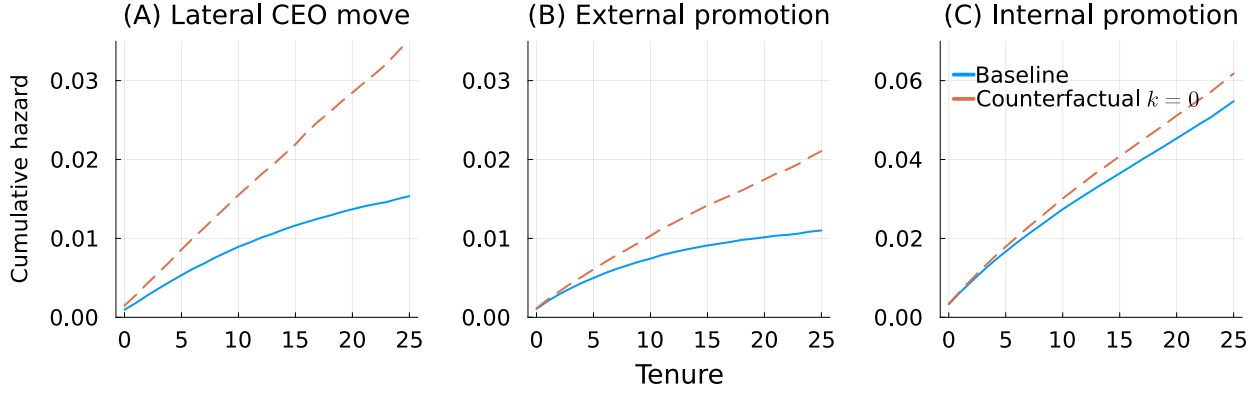
The previous section highlights the immense influence of competition on the level of CEO pay. This result may be somewhat surprising; the rigidity featured in the market for executives may seem to suggest that the degree of competition is minimal. In other words, if the overwhelming majority of CEOs work for only one firm over their career, how is it that across-firm competition has such an enduring impact on their compensation? This apparent puzzle is the focal point of this section, which we resolve by illustrating the subtle effects of firm-specific human capital on CEOs' ability to extract rents. To this end, we compare CEO compensation and employment outcomes in the estimated model to a counterfactual version of the model which eliminates firm-specific human capital entirely. We first discuss the extent to which executive labor market rigidity is explained by imperfect portability of human capital. We then examine how managerial bargaining outcomes, and the persistent effect of competition, are affected by their human capital. Though firm-specific human capital hampers managerial mobility, we find that it notably enhances their capacity for rent extraction.

7.1. Human Capital and Transitions into CEO Employment

We first discuss the effect of firm-specific human capital accumulation on across-firm mobility in the managerial labor market. In Figure 3, we compare the baseline and counterfactual probabilities of transitioning into CEO employment through a lateral move, external promotion, or internal promotion. When human capital is partially-firm specific, there is a marked decline in the rate of across-firm transitions, whether a lateral CEO move or external promotion. This is the

Figure 3. Firm-specific human capital and transition rates

This figure displays cumulative hazard rates of a manager experiencing (A) a lateral CEO move (conditional on being a CEO); (B) an external promotion (conditional on being an executive); and (C) an internal promotion (conditional on being an executive). The blue line displays the estimated rates, the orange lines the counterfactual when $k = 0$.



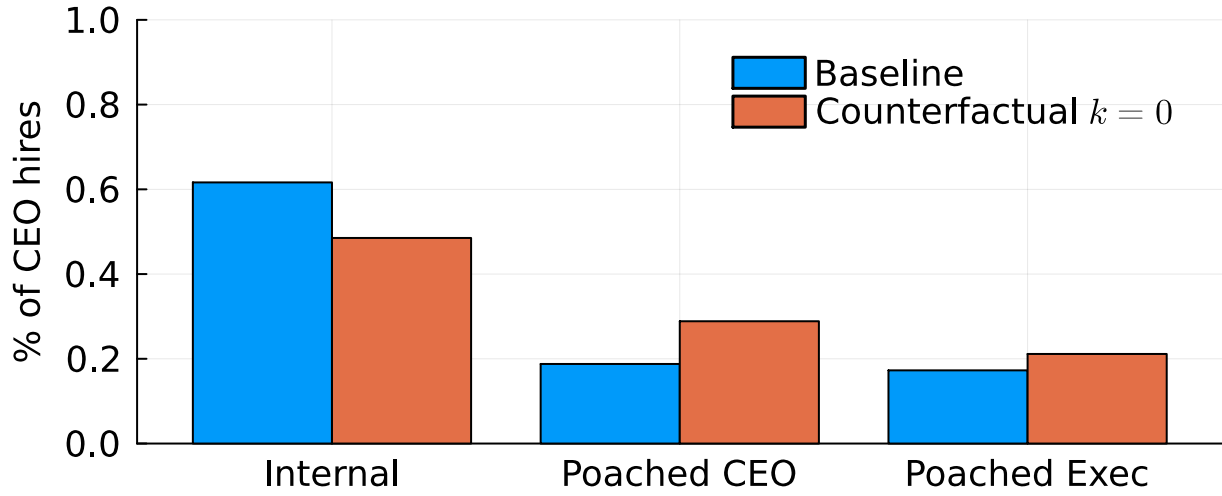
result of job lock; as firm-specific human capital accumulates, the opportunity cost of switching employers gradually increases. However, even in the absence of firm-specific human capital, the rate of across-firm transitions is still quite low. The counterfactual rate of lateral CEO moves, for example, is only roughly 3%.

In contrast with the across-firm transition rates the rate of internal promotions is virtually unaffected by firm-specific human capital. This may come at no surprise, as managerial capital is fully retained upon switching positions within a given firm. Though this rate is stable, the increased frequency of across-firm transitions will imply that a smaller share of CEOs are appointed from within the firm when firm-specific human capital is immaterial. We illustrate this in Figure 4, which plots the baseline and counterfactual shares of CEOs hired from within the firm, poached from CEO positions, or poached from executive positions. As expected, the proportion of internally-hired CEOs decreases when eliminating firm-specific capital while the share of external hires increases. Clearly, internal promotions remain the most common route to CEO employment even in the counterfactual case. Thus, while firm-specific human capital accumulation explains a portion of firms' "preference" for hiring internally, it is clear that other factors contribute significantly to this hiring disparity.

The remaining gap in rates of CEO appointments is attributable to a combination of exogenous and endogenous characteristics of the executive labor market. In our model, the key exogenous

Figure 4. Firm-specific human capital and internal vs. external CEO hiring

This figure displays baseline and counterfactual (with $k = 0$) proportions of internal CEO promotions, externally poached CEOs and externally poached non-CEO executives.

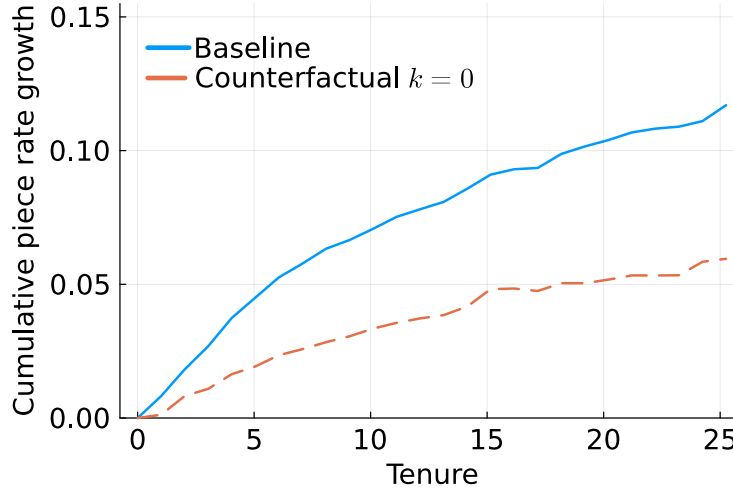


explanatory features are the disparate rates of job offers across managerial types. As previously discussed, our estimates of the arrival rates for internal and external CEO offers (λ_0 and λ_1) imply that insiders are significantly more likely to receive offers in the first place. As these parameters are exogenously given in our model, whether this gap in arrival rates is the product of inefficient favoritism for internal candidates or the efficient bypassing of external search costs is unclear and outside the scope of our analysis. We can be sure, however, that this disparity in hiring rates is due in large part to endogenous labor market competition.

When attempting to poach a manager, whether a CEO or executive, firms contend not only with the manager during negotiations, but also with the manager's incumbent employer. The competition that results decreases the likelihood of a successful poach, as incumbent employers have the capacity to dissuade managerial exit via contractual renegotiations. As mentioned previously, negotiations with internal CEO candidates are free from such competitive pressure and are thus more likely to result in a successful CEO appointment. In this sense, internal CEO promotions represent a "path of least resistance" in which firms can avoid confrontations with a competing firm. Hence, the high share of internal CEO hires is not entirely the product of labor market frictions. This outcome is to be expected in the face of well-functioning labor market competition.

Figure 5. Firm-specific human capital and CEO rent extraction

This figure displays how the CEO's accumulated piece growth changes over tenure, both in the main simulated model and in a counterfactual where firm-specific human capital is zero.



7.2. Firm-Specific Human Capital and CEO Rent Extraction

The previous discussion highlights the following, somewhat counterintuitive, result: the lack of across-firm CEO movement is in part the consequence of across-firm competition. Firms' defensive bidding tactics decrease the rate of CEO transitions while simultaneously increasing the level of CEO pay. These bidding tactics take the form of renegotiations in our model, and are the key channel through which CEOs can dynamically increase their share of rents within a given position. As discussed in Section 3, conditional on receiving a competing job offer, the likelihood of contractual renegotiations is strictly higher when human capital is partially firm specific. This is effectively the result of job lock; firm-specific human capital gradually increases CEOs' acceptance threshold, decreasing poachers' ability to outbid the incumbent firms. Rather than switch positions, CEOs elect to remain with their employers under a renegotiated contract. In Figure 5, we compare average within-position piece rate growth in the baseline and counterfactual versions of our model.

Firm-specific human capital roughly doubles the rate at which CEO piece rates grow with tenure. Again, this is due to the fact that renegotiations, as opposed to transitions, are more likely when human capital has a firm-specific component. This illustrates that despite the fact that firm-

specific capital dampens mobility (Figure 3), it magnifies the beneficial impact of competition on within-job compensation growth. Firm-specific human capital weakens *offensive* bidding tactics by poaching firms while strengthening *defensive* bidding tactics by incumbent firms. Put together, this enhances CEOs’ capacity to extract surplus from their firms and positively impacts the level of CEO compensation.

8. Conclusion

This paper presents a structural model of the executive labor market that incorporates general and firm-specific capital accumulation, managerial bargaining power, search frictions and labor market competition, internal and external promotions, and firm and executive heterogeneity. Our model reveals theoretically that realized CEO surplus capture is path-dependent: poached CEOs capture more rents than poached non-CEO executives and internally promoted CEOs.

Our paper makes two primary empirical contributions. First, we measure the relative importance of general and firm-specific human capital in managerial skill over the manager’s career, and show that firm-specific human capital lowers cross-firm mobility (less job-switching), but raises within-firm mobility (higher surplus capture with current firm). Second, we decompose realized CEO surplus capture into contributions from pure CEO bargaining power and labor market competition and show that labor market competition is generally the larger component.

Our work has important implications for the corporate finance literature. First, our decomposition of managerial human capital is crucial for understanding the CEO labor market, particularly the literature focusing on general CEO skill (Gabaix and Landier, 2008; Murphy and Zabojnik, 2007; Cziraki and Jenter, 2024; Graham et al., 2020a). To the best of our knowledge, our paper is the first in the literature to quantify directly the composition of managerial human capital and then the impact of firm-specific skill on the managerial contract.

Second, our decomposition of realized CEO surplus is important for understanding the role of CEO bargaining power and agency frictions in determining CEO compensation. We show that, even in an imperfect labor market with search frictions, in which CEO-CEO transitions are rare,

CEO surplus capture can largely be explained by competition in the labor market. In models without an explicit managerial labor market, estimates of agency frictions which induce high CEO wages or low turnover may be over-stated.

References

- Bagger, J., Fontaine, F., Postel-Vinay, F., and Robin, J.-M. (2014). Tenure, experience, human capital, and wages: A tractable equilibrium search model of wage dynamics. *The American Economic Review*, 104(6):1551–1596.
- Barry, J. W. (2023). Shareholder voice and executive compensation. *Available at SSRN 4584580*.
- Bebchuk, L. A., Cremers, K. M., and Peyer, U. C. (2011). The ceo pay slice. *Journal of financial Economics*, 102(1):199–221.
- Bebchuk, L. A. and Fried, J. M. (2003). Executive compensation as an agency problem. *Journal of economic perspectives*, 17(3):71–92.
- Bebchuk, L. A. and Fried, J. M. (2006). Pay without performance: Overview of the issues. *Academy of Management Perspectives*, 20(1):5–24.
- Bertrand, M. and Mullainathan, S. (2001). Are ceos rewarded for luck? the ones without principals are. *The Quarterly Journal of Economics*, 116(3):901–932.
- Cahuc, P., Postel-Vinay, F., and Robin, J.-M. (2006). Wage bargaining with on-the-job search: Theory and evidence. *Econometrica*, 74(2):323–364.
- Capron, A., Hamilton, B., Hincapié, A., and Lyman, N. (2024). Buying the lottery in bad times: Why do firms hire outside ceos? *Working Paper*.
- Coles, J. L., Daniel, N. D., and Naveen, L. (2014). Co-opted boards. *The Review of Financial Studies*, 27(6):1751–1796.
- Core, J. E., Holthausen, R. W., and Larcker, D. F. (1999). Corporate governance, chief executive officer compensation, and firm performance. *Journal of financial economics*, 51(3):371–406.
- Custódio, C., Ferreira, M. A., and Matos, P. (2013). Generalists versus specialists: Lifetime work experience and chief executive officer pay. *Journal of Financial Economics*, 108(2):471–492.
- Cziraki, P. and Jenter, D. (2024). The market for ceos. *Working Paper*.
- Edmans, A., Gabaix, X., and Landier, A. (2009). A multiplicative model of optimal ceo incentives in market equilibrium. *The Review of Financial Studies*, 22(12):4881–4917.
- Fahlenbrach, R. (2009). Shareholder rights, boards, and ceo compensation. *Review of Finance*, 13(1):81–113.
- Frydman, C. (2019). Rising through the ranks: The evolution of the market for corporate executives, 1936–2003. *Management Science*, 65(11):4951–4979.
- Gabaix, X. and Landier, A. (2008). Why has ceo pay increased so much? *The quarterly journal of economics*, 123(1):49–100.

- Gao, J., Wang, W., and Wu, Y. (2021). Human capital portability and worker career choices: Evidence from m&a bankers. *Available at SSRN 3724657*.
- Geelen, T. and Hajda, J. (2024). Succession. *Working paper*.
- Graham, J. R., Grennan, J., Harvey, C. R., and Rajgopal, S. (2022). Corporate culture: Evidence from the field. *Journal of financial economics*, 146(2):552–593.
- Graham, J. R., Kim, D., and Kim, H. (2020a). Executive mobility in the united states, 1920 to 2011. *Working Paper*.
- Graham, J. R., Kim, H., and Leary, M. (2020b). Ceo-board dynamics. *Journal of Financial Economics*, 137(3):612–636.
- He, L. and Schroth, E. J. (2024). Why are ceos rarely succeeded by external candidates? *Available at SSRN 4728228*.
- Lazear, E. P. (2009). Firm-specific human capital: A skill-weights approach. *Journal of political economy*, 117(5):914–940.
- Ma, S., Wang, W., and Wu, Y. (2023). Steering labor mobility through innovation. *Working Paper*.
- Mincer, J. (1974). Schooling, experience, and earnings. *Human Behavior & Social Institutions* No. 2.
- Morse, A., Nanda, V., and Seru, A. (2011). Are incentive contracts rigged by powerful ceos? *The Journal of Finance*, 66(5):1779–1821.
- Murphy, K. J. and Zabojnik, J. (2004). Ceo pay and appointments: A market-based explanation for recent trends. *American economic review*, 94(2):192–196.
- Murphy, K. J. and Zabojnik, J. (2007). Managerial capital and the market for ceos. *Available at SSRN 984376*.
- Page, T. B. (2018). Ceo attributes, compensation, and firm value: Evidence from a structural estimation. *Journal of Financial Economics*, 128(2):378–401.
- Postel-Vinay, F. and Robin, J.-M. (2002). Equilibrium wage dispersion with worker and employer heterogeneity. *Econometrica*, 70(6):2295–2350.
- Rubinstein, A. (1982). Perfect equilibrium in a bargaining model. *Econometrica: Journal of the Econometric Society*, pages 97–109.
- Shi, L. (2023). Optimal regulation of noncompete contracts. *Econometrica*, 91(2):425–463.
- Smith, A. A. (2016). *Indirect Inference*, pages 1–6. Palgrave Macmillan UK, London.
- Taylor, L. A. (2010). Why are ceos rarely fired? Evidence from structural estimation. *The Journal of Finance*, 65(6):2051–2087.

- Taylor, L. A. (2013). Ceo wage dynamics: Estimates from a learning model. *Journal of Financial Economics*, 108(1):79–98.
- Terviö, M. (2008). The difference that ceos make: An assignment model approach. *American Economic Review*, 98(3):642–668.

Table 1. Summary and labor market transition statistics

This table displays summary statistics for managerial wages, and labor market experience and tenure (Panel A); and labor market transition statistics (Panel B). In Panel A, statistics are displayed based on employment status (executive or CEO). In Panel B, labor market transition statistics are displayed conditional on employment state (executive, CEO, unattached); the first two columns display the number of transitions and the proportion of observations conditional on state. Columns 3 and 4 display the average years of experience or tenure that the transition occurs, respectively.

Panel A: Summary statistics

	N	Mean	Std Dev	25%	50%	75%
CEOs						
Log wage	60,104	14.859	1.209	14.071	14.919	15.687
Wage (\$ million)	60,104	5.351	10.943	1.291	3.016	6.497
Wage Growth (%)	55,188	10.420	62.775	-14.997	7.436	35.380
Experience	60,104	22.105	11.583	13	21	31
Tenure	60,056	15.254	11.372	6	12	22
Non-CEO executives						
Log wage	263,729	13.791	1.070	13.051	13.773	14.508
Wage (\$ million)	263,729	1.748	3.915	0.466	0.959	1.999
Wage Growth (%)	211,378	13.457	58.116	-11.005	9.390	36.543
Experience	263,729	15.251	11.916	5	12	23
Tenure	262,659	5.780	5.877	2	4	7

Panel B: Labor market transition statistics

	(1) N	(2) %	(3) Experience	(4) Tenure
CEOs				
Horizontal CEO move	246	0.409	21.626	12.098
Unattachment	530	0.882	20.517	11.483
Retirement	8,687	14.453	23.607	15.523
Non-CEO executives				
Internal CEO promotion	4,568	1.732	16.914	9.959
External CEO promotion	329	0.125	16.581	5.854
Horizontal executive move	1,574	0.597	14.386	4.810
Unattachment	5,109	1.937	15.062	4.611
Retirement	41,215	15.628	15.689	6.344
Unattached managers				
Executive position	5,208	23.216	19.253	
CEO position	698	3.111	22.232	

Table 2. Structural parameter estimates

This table displays the estimated structural parameters, given the estimation sample describe in Table 1.

μ	η	λ_0	λ_1	λ_2	γ_E	γ_C	β	σ_a	p_{min}	s_1	s_2	δ_g	δ_k
0.038	0.040	0.156	0.084	0.009	0.132	0.988	0.136	0.439	12.7	1.88	1.57	0.032	0.036

Table 3. CEO human capital composition across experience and tenure

This table shows a decomposition of accumulated CEO human capital into general human capital $g(t)$, and firm-specific human capital $k(\tau)$. Panel A displays the decomposition across labor market experience t and Panel B does so for firm tenure. We compute shares in levels: $s_x = \frac{\exp(x)}{\exp(g)+\exp(k)}$ for each $x \in \{g, k\}$.

Panel A: Human capital decomposition by labor market experience

	(1)	(2)	(3)	(4)	(5)
	Full sample	Conditional on experience (t)			
		[0, 5)	[5, 10)	[10, 25)	≥ 25
General human capital (g)	54.36%	50.14%	51.19%	54.16%	59.43%
Firm-specific human capital (k)	45.64%	49.86%	48.81%	45.84%	40.57%

Panel B: Human capital decomposition by firm tenure

	(1)	(2)	(3)	(4)	(5)
	Full sample	Conditional on experience (t)			
		[0, 5)	[5, 10)	[10, 25)	≥ 25
General human capital (g)	54.36%	55.22%	54.96%	53.93%	50.36%
Firm-specific human capital (k)	45.64%	44.78%	45.04%	46.07%	49.64%

Table 4. CEO surplus capture decomposition

This table shows realized CEO surplus capture. Panel A displays realized CEO surplus capture by eventual CEO hire type. Panel B displays across labor market experience and Panel B by firm tenure. We display realized CEO surplus capture $\mathbb{E}[b \mid X]$, for comparison against exogenously estimated pure CEO bargaining power $\beta_1 = 13.6\%$ in (27). The first row displays average realized surplus capture b . The second row displays the share attributable to labor market competition

Panel A: CEO surplus capture decomposition by CEO hire type

	(1)	(2)	(3)	(4)
	Full sample	Conditional on CEO hire type		
		Internal	Poached CEO	Poached exec
Realized surplus capture (b)	56.35%	51.54%	68.64%	64.22%
Labor market competition ($\frac{b-\beta_1}{b}$)	75.94%	73.70%	80.25%	78.89%

Panel B: CEO surplus capture decomposition by labor market experience

	(1)	(2)	(3)	(4)	(5)
	Full sample	Conditional on experience (t)			
		[0, 5)	[5, 10)	[10, 25)	≥ 25
Realized surplus capture (b)	56.35%	24.23%	52.57%	63.59%	65.45%
Labor market competition ($\frac{b-\beta_1}{b}$)	75.94%	44.05%	74.21%	78.68%	79.29%

Panel C: CEO surplus capture decomposition by firm tenure

	(1)	(2)	(3)	(4)	(5)
	Full sample	Conditional on tenure (τ)			
		[0, 5)	[5, 10)	[10, 25)	≥ 25
Realized surplus capture (b)	56.35%	42.47%	55.64%	61.66%	61.59%
Labor market competition ($\frac{b-\beta_1}{b}$)	75.94%	68.08%	75.64%	78.01%	77.99%

A. Model Appendix

A.1. Derivation of Bargaining Rules

Derivation of sharing rule for internal promotions. Suppose an executive is approached by their firm and evaluated for a promotion to CEO. First, a new match productivity parameter p' is drawn, summarizing the executive's fitness for the CEO position. Each party makes alternating offers over the piece rate r' . If the offer is accepted, the bargaining game ends. If the offer is rejected, some time elapses before a counteroffer is made. Let Δ_e and Δ_f respectively denote the lengths of time which elapse following a rejection by the executive and firm. It is also assumed that during negotiations, the match severs at rate η in which case the executive remains in their current position, and additional offers for outside CEO and non-CEO positions respectively arrive at rates λ_1 and λ_2 . The subgame perfect equilibrium of this game consists of piece rate offers (r_e, r_f) which make the other party indifferent between immediate acceptance and waiting to make a counteroffer. That is, r_e and r_f respectively solve:

$$V_C(r_f, h, \tau, p') = \frac{1}{1 + \rho\Delta_e} \left[w_t\Delta_e + \eta\Delta_e V_E(r, h, \tau, p) + \lambda_1\Delta_e \tilde{V}_C(\cdot) + \lambda_2\Delta_e \tilde{V}_E(\cdot) + (1 - \Delta_e(\eta + \lambda_1 + \lambda_2))V_C(r_e, h, \tau, p') \right] \quad (\text{A.1})$$

$$\Pi_C(r_e, h, \tau, p') = \frac{1}{1 + \rho\Delta_f} \left[\pi_0\Delta_e + \eta\Delta_e \Pi_0 + \lambda_0\Delta_f \tilde{\Pi}_C(\cdot) + \lambda_1\Delta_f \tilde{\Pi}_C(\cdot) + (1 - \Delta_f(\eta + \lambda_0 + \lambda_1))\Pi_C(r_f, h, \tau, p') \right] \quad (\text{A.2})$$

$\Pi_C(x)$ denotes the value to the firm of filling the CEO position given state x . π_0 and Π_0 denote the flow and net present values to the firm of having a vacant CEO position, both of which we assume to equal 0. $\tilde{V}_C$ and $\tilde{V}_E$ denote the executive's net present value of initiating a new bargaining game for a CEO or non-CEO position upon the arrival of a competing offer. Similar for $\tilde{\Pi}_C$. The two equations above can be rewritten as:

$$V_C(r_f, h, \tau, p') - V_C(r_e, h, \tau, p') = -\Delta_e \left[(\eta + \lambda_1 + \lambda_2)V_C(r_e, h, \tau, p') + \rho V_C(r_f, h, \tau, p') \right]$$

$$- w_t - \eta V_E(r, h, \tau, p) - \lambda_1 \tilde{V}_C(\cdot) - \lambda_2 \tilde{V}_E(\cdot) \Big] \quad (\text{A.3})$$

$$\begin{aligned} \Pi_C(r_e, h, \tau, p') - \Pi_C(r_f, h, \tau, p') = & -\Delta_f \left[(\eta + \lambda_0 + \lambda_1) \Pi_C(r_f, h, \tau, p') + \rho \Pi_C(r_e, h, \tau, p') \right. \\ & \left. - \pi_0 - \eta \Pi_0 - \lambda_0 \tilde{\Pi}_C(\cdot) - \lambda_1 \tilde{\Pi}_E(\cdot) \right] \end{aligned} \quad (\text{A.4})$$

The above conditions imply that $r_f \rightarrow r_e$ as $\Delta_f, \Delta_e \rightarrow 0$. Denote their common limit by r' and define:

$$\frac{\partial V_C}{\partial r}(r, h, \tau, p') = \lim_{\Delta_f, \Delta_e \rightarrow 0} \frac{V_C(r_f, h, \tau, p') - V_C(r_e, h, \tau, p')}{r_f - r_e} \quad (\text{A.5})$$

$$\frac{\partial \Pi_C}{\partial r}(r, h, \tau, p') = \lim_{\Delta_f, \Delta_e \rightarrow 0} \frac{\Pi_C(r_f, h, \tau, p') - \Pi_C(r_e, h, \tau, p')}{r_f - r_e} \quad (\text{A.6})$$

Using the definitions above and taking the ratios of (A.3) and (A.4) yields:

$$-\frac{\frac{\partial V_C}{\partial r}(r', h, \tau, p')}{\frac{\partial \Pi_C}{\partial r}(r', h, \tau, p')} = \frac{\Delta_e(\rho + \eta + \lambda_1 + \lambda_2)}{\Delta_f(\rho + \eta + \lambda_0 + \lambda_1)} \frac{V_C(r', h, \tau, p') - \frac{w_t + \eta V_E(r, h, \tau, p) + \lambda_1 \tilde{V}_C(\cdot) + \lambda_2 \tilde{V}_E(\cdot)}{\rho + \eta + \lambda_1 + \lambda_2}}{\Pi_C(r', h, \tau, p') - \frac{\pi_0 + \eta \Pi_0 + \lambda_0 \tilde{\Pi}_C(\cdot) + \lambda_1 \tilde{\Pi}_E(\cdot)}{\rho + \eta + \lambda_0 + \lambda_1}} \quad (\text{A.7})$$

Next, define $S(h, \tau, p') = \Pi_C(r', h, \tau, p') + V_C(r', h, \tau, p') - V_E(r, h, \tau, p)$ as the surplus associated with the position. Note that $\Pi_C(0, h, \tau, p') = 0$, which implies that $\Pi_C(r', h, \tau, p') = V_C(0, h, \tau, p') - V_C(r', h, \tau, p')$. Thus, $\frac{\partial \Pi_C}{\partial r}(r', h, \tau, p') = -\frac{\partial V_C}{\partial r}(r', h, \tau, p')$. Applying this to (A.7) and taking the limit as $\eta \rightarrow \infty$ yields (after some algebra):

$$V_C(r', h, \tau, p') = \beta_0 V_C(0, h, \tau, p') + (1 - \beta_0) V_E(r, h, \tau, p) \quad (\text{A.8})$$

where $\beta_0 = \frac{\Delta_f}{\Delta_f + \Delta_e}$.¹ Note that the sharing rule implies that it is in the executive's interest to accept the promotion to CEO if and only if $V_C(0, h', \tau', p') > V_E(r, h', \tau', p)$. Define $\bar{\theta}_0(r, h, \tau, p)$ as the

¹Under this game-theoretic interpretation of the bargaining parameter, the executive's bargaining power increases in the time Δ_f it takes the firm to come up with a counteroffer. This is fairly unimportant and in no way changes the empirical treatment of the parameter β_0 . Also, the form of the sharing rules hinges on the assumption that the separation rate η is large relative to the arrival rates of competing offers. This is a technical assumption that for our purposes will be buried in the appendix.

critical level of match productivity such that:

$$V_C(0, h, \tau, \bar{\theta}_0) = V_E(r, h, \tau, p) \quad (\text{A.9})$$

If $p' > \bar{\theta}_0$, the executive accepts the CEO role under the wage contract defining (A.8). If $p' \leq \bar{\theta}_0$, the executive can never be swayed into accepting the CEO role and instead remains in their current position.

Derivation of sharing rule for horizontal moves (Executive). Consider an executive who is approached by an outside firm to serve in an executive position. Upon the arrival of the offer, the executive along with the competing and incumbent firms initiate a bargaining game with the following structure:

1. Stage 1: Both firms simultaneously offer a piece rate to the executive
2. Stage 2: The executive chooses one of the offers, or rejects and keeps their current position.
3. Stage 3: If the executive accepted an offer in Stage 2, some time elapses. The executive then renegotiates with the firm whose offer was rejected, where the renegotiation protocol mirrors that of the previous section. Unlike the previous section, however, the executive's outside option is not their current position, but the offer accepted in Stage 2.

The bargaining game is solved via backward induction. Let r'_1 and r_1 denote the Stage 1 offers from firms p' and p . Suppose that r_1 was accepted in the second stage, triggering a Stage 3 renegotiation with firm p' . The offer from p' will satisfy:

$$V_E(r, h, 0, p') = \beta_1 V_E(0, h, 0, p') + (1 - \beta_1) V_E(r_1, h, \tau, p) \quad (\text{A.10})$$

Conversely, suppose that r'_1 was accepted in Stage 2. In the subsequent renegotiation with firm p , their counteroffer will satisfy:

$$V_E(r, h, \tau, p) = \beta_1 V_E(0, h, \tau, p) + (1 - \beta_1) V_E(r'_1, h, 0, p') \quad (\text{A.11})$$

The form of the counteroffers implies that:

- If r'_1 was accepted in Stage 2, renegotiate and eventually work with p iff:

$$V_E(0, h, \tau, p) \geq V_E(r'_1, h, 0, p') \quad (\text{A.12})$$

- If r_1 was accepted in Stage 2, renegotiate and eventually work with p' iff:

$$V_E(0, h, 0, p') > V_E(r_1, h, \tau, p) \quad (\text{A.13})$$

Thus, the value of accepting r_1 at Stage 2 is:

$$V = \max\{\beta_1 V_E(0, h, 0, p') + (1 - \beta_1) V_E(r_1, h, \tau, p), V_E(r_1, h, \tau, p)\} \quad (\text{A.14})$$

Similarly for r'_1 :

$$V = \max\{\beta_1 V_E(0, h, \tau, p) + (1 - \beta_1) V_E(r'_1, h, 0, p'), V_E(r'_1, h, 0, p')\} \quad (\text{A.15})$$

Moving back to Stage 1, both firms make simultaneous offers. For firm p' to eventually win the executive, they must bid r'_1 such that $V_E(r'_1, h, 0, p') > V_E(0, h, \tau, p)$ so that firm p cannot afford to outbid p' . Acknowledging the transferability of p and k , firm p' eventually wins the worker if and only if $p' > p + k'$. In this case, to avoid wasting time in the renegotiation stage, firm p' immediately offers r'_1 such that:

$$V_E(r'_1, h, 0, p') = \beta_1 V_E(0, h, 0, p') + (1 - \beta_1) V_E(0, h, \tau, p) \quad (\text{A.16})$$

Conversely, if $p' < p + k'$, firm p will eventually win the executive's services. The fastest way of doing so is to immediately offer r_1 such that:

$$V_E(r_1, h, \tau, p) = \beta_1 V_E(0, h, \tau, p) + (1 - \beta_1) V_E(0, h, 0, p') \quad (\text{A.17})$$

Note additionally that a competing offer does not necessitate an alteration of the current piece rate r . The minimal value of p' such that something happens is defined by:

$$V_E(r, h, \tau, p) = \beta_1 V_E(0, h, \tau, p) + (1 - \beta_1) V_E(0, h, 0, \underline{\theta}_E) \quad (\text{A.18})$$

Derivation of sharing rule for external promotions. Consider an executive who is approached by an outside firm to become CEO. A three-player bargaining game is initiated with the same structure as in the previous case. Note, however, that unlike the previous case, the executive is weighing two separate types of positions: a CEO position and a non-CEO position. Because the two position types are associated with different event spaces describing the possible set of future offers, the executive's value of accepting these positions, for a given state, is not equal in general.

As before, the bargaining game is solved via backward induction. Let $(r'_1, 1)$ and $(r_1, 0)$ denote the stage 1 offers from firms p' and p , where the second coordinate indicates if the offer is for a CEO position or not. Suppose the executive accepted $(r_1, 0)$ at Stage 2, then renegotiates with p' in Stage 3. The offer from p' will satisfy:

$$V_C(r, h, 0, p') = \beta_1 V_C(0, h, 0, p') + (1 - \beta_1) V_E(r_1, h, \tau, p) \quad (\text{A.19})$$

Conversely, suppose that $(r'_1, 1)$ was accepted at stage 2, and a stage 3 renegotiation was triggered with firm p . Firm p 's counteroffer will satisfy:

$$V_E(r, h, \tau, p) = \beta_1 V_E(0, h, \tau, p) + (1 - \beta_1) V_C(r'_1, h, 0, p') \quad (\text{A.20})$$

Implications:

- If $(r'_1, 1)$ is accepted at stage 2, renegotiate and eventually work with p iff:

$$V_E(0, h, \tau, p) \geq V_C(r'_1, h, 0, p') \quad (\text{A.21})$$

- If $(r_1, 0)$ is accepted at stage 2, renegotiate and eventually work with p' iff:

$$V_C(0, h, 0, p') > V_E(r_1, h, \tau, p) \quad (\text{A.22})$$

Thus, the value of accepting $(r_1, 0)$ at stage 2 is:

$$V = \max\{\beta_1 V_C(0, h, 0, p') + (1 - \beta_1) V_E(r_1, h, \tau, p), V_E(r_1, h, \tau, p)\} \quad (\text{A.23})$$

Similarly for $(r'_1, 1)$:

$$V = \max\{\beta_1 V_E(0, h, \tau, p) + (1 - \beta_1) V_C(r_1, h, 0, p), V_C(r'_1, h, 0, p')\} \quad (\text{A.24})$$

At stage 1, simultaneous offers are made. For firm p' to win the executive, they must bid r'_1 such that: $V_C(r'_1, h, 0, p') > V_E(0, h, \tau, p) = V_C(0, h, \tau, \bar{\theta}_0)$. Again acknowledging the transferability of k and p , the previous inequality is equivalent to $V_C(r'_1, h', 0, p') > V_C(0, h', 0, \bar{\theta}_0 + k')$. Hence, firm p' eventually wins the worker if and only if $p' > \bar{\theta}_0 + k'$. In this case, to avoid wasting time in the renegotiation stage, firm p' immediately offers r'_1 such that:

$$V_C(r'_1, h, 0, p') = \beta_1 V_C(0, h, 0, p') + (1 - \beta_1) V_E(0, h, \tau, p) \quad (\text{A.25})$$

Conversely, suppose that $p' < \bar{\theta}_0 + k'$. Similar to the above case, firm p will retain the worker in the fastest manner possible by immediately offering r_1 such that:

$$V_E(r_1, h, \tau, p) = \beta_1 V_E(0, h, \tau, p) + (1 - \beta_1) V_C(0, h, 0, p') \quad (\text{A.26})$$

As in the case of horizontal moves, an outside offer for a CEO appointment need not trigger a change in the current piece rate r . The minimum value of p' such that something happens is defined by:

$$V_E(r, h, \tau, p) = \beta_1 V_E(0, h, \tau, p) + (1 - \beta_1) V_C(0, h, 0, \underline{\theta}_O) \quad (\text{A.27})$$

Derivation of sharing rule for horizontal moves (CEO). This case is identical to the case for horizontal executive moves if we simply change subscripts. Upon receiving an offer for a CEO position with match productivity $p' > p+k$, the CEO switches positions and receives initial piece rate r' defined by condition:

$$V_E(r', h, \tau, p) = \beta_1 V_E(0, h, \tau, p) + (1 - \beta_1) V_C(0, h, 0, p') \quad (\text{A.28})$$

As in the previous cases, if instead $p' > p+k$, the current firm retains the CEO and a renegotiation may be triggered. The minimum value of p' such that the piece rate is revised is defined by:

$$V_C(r, h, \tau, p) = \beta_1 V_C(0, h, \tau, p) + (1 - \beta_1) V_C(0, h, 0, \underline{\theta}_C) \quad (\text{A.29})$$

A.2. CEO Value Function Derivation

To obtain closed-form expressions for CEO piece rates, we first derive a simplified version of the value function (12). Rearranging terms via integration by parts yields:

$$\begin{aligned} (\rho + \mu + \eta) V_C(r, k, p) &= r + g + k + p + \eta V_U(h) + \lambda_1 \beta_1 \int_{p+k}^{p_{\max}} \frac{\partial V_C}{\partial x}(0, 0, x) S(x) dx \\ &\quad + \lambda_1 (1 - \beta_1) \int_{\underline{\theta}_C}^{p+k} \frac{\partial V_C}{\partial x}(0, 0, x) S(x) dx \end{aligned} \quad (\text{A.30})$$

Setting $r = 0$ and differentiating with respect to p (applying Leibniz' rule) yields:

$$\frac{\partial V_C}{\partial p}(0, k, p) = \frac{1}{\rho + \mu + \eta + \lambda_1 \beta_1 S(p)} \equiv \phi_C(p) \quad (\text{A.31})$$

The value function (12) can then be expressed as:

$$\begin{aligned}
(\rho + \mu + \eta)V_C(r, k, p) &= r + g + k + p + \eta V_U(h) + \lambda_1 \beta_1 \int_{p+k}^{p_{max}} \phi_C(x) S(x) dx \\
&\quad + \lambda_1 (1 - \beta_1) \int_{\underline{\theta}_C}^{p+k} \phi_C(x) S(x) dx
\end{aligned} \tag{A.32}$$

A.3. Executive Value Function Derivation

Similar to the case of CEOs, we simplify the executive value function by first rearranging (13) via integration by parts:

$$\begin{aligned}
(\rho + \mu + \eta)V_E(r, k, p) &= r + g + k + p + \eta V_U(h) + \lambda_0 \beta_0 \int_{\bar{\theta}}^{p_{max}} \frac{\partial V_C}{\partial x}(0, k, x) S(x) dx \\
&\quad + \lambda_1 \beta_1 \int_{\bar{\theta}+k}^{p_{max}} \frac{\partial V_C}{\partial x}(0, 0, x) S(x) dx + \lambda_1 (1 - \beta_1) \int_{\underline{\theta}_O}^{\bar{\theta}+k} \frac{\partial V_C}{\partial x}(0, 0, x) S(x) dx \\
&\quad + \lambda_2 \beta_1 \int_{p+k}^{p_{max}} \frac{\partial V_E}{\partial x}(0, 0, x) S(x) dx + \lambda_2 (1 - \beta_1) \int_{\underline{\theta}_E}^{p+k} \frac{\partial V_E}{\partial x}(0, 0, x) S(x) dx
\end{aligned} \tag{A.33}$$

Again setting $r = 0$ and differentiating with respect to p yields:

$$\frac{\partial V_E}{\partial p}(0, k, p) = \frac{1 - \phi_C(\bar{\theta} + k) \frac{\partial \bar{\theta}}{\partial p} \left(\lambda_0 \beta_0 S(\bar{\theta}) + \lambda_1 \beta_1 S(\bar{\theta} + k) \right)}{\rho + \mu + \eta + \lambda_2 \beta_1 S(p + k)} \tag{A.34}$$

Differentiating Equation (5) evaluated at $r = 0$ yields:

$$\frac{\partial \bar{\theta}}{\partial p}(0, k, p) = \frac{\frac{\partial V_E}{\partial p}(0, k, p)}{\phi_C(\bar{\theta} + k)} \tag{A.35}$$

Substituting the expression above into Equation (A.34) and simplifying then yields:

$$\frac{\partial V_E}{\partial p}(0, k, p) = \frac{1}{\rho + \mu + \eta + \lambda_2 \beta_1 S(p + k) + \lambda_1 \beta_1 S(\bar{\theta} + k) + \lambda_0 \beta_0 S(\bar{\theta})} \equiv \phi_E(p, k) \tag{A.36}$$

The executive value function can then be expressed as:

$$\begin{aligned}
(\rho + \mu + \eta)V_E(r, k, p) = & r + g + k + p + \eta V_U(h) + \lambda_0 \beta_0 \int_{\bar{\theta}}^{p_{max}} \phi_C(x + k)S(x)dx \\
& + \lambda_1 \beta_1 \int_{\bar{\theta}+k}^{p_{max}} \phi_C(x)S(x)dx + \lambda_1(1 - \beta_1) \int_{\underline{\theta}_O}^{\bar{\theta}+k} \phi_C(x)S(x)dx \\
& + \lambda_2 \beta_1 \int_{p+k}^{p_{max}} \phi_E(x, 0)S(x)dx + \lambda_2(1 - \beta_1) \int_{\underline{\theta}_E}^{p+k} \phi_E(x, 0)S(x)dx \quad (A.37)
\end{aligned}$$

With both the executive and CEO value functions in hand, we can present the formal proof of Theorem 1.

Proof of Theorem 1. Observe that for all values of p :

$$\frac{\partial \bar{\theta}}{\partial p}(0, k, p) = \frac{\phi_C(p, k)}{\phi_E(\bar{\theta} + k)} = \frac{\rho + \mu + \eta + \lambda_1 \beta_1 \bar{F}(\bar{\theta} + k)}{\rho + \mu + \eta + \lambda_2 \beta_1 \bar{F}(p + k) + \lambda_1 \beta_1 \bar{F}(\bar{\theta} + k) + \lambda_0 \beta_0 \bar{F}(\bar{\theta})} \leq 1 \quad (A.38)$$

Then:

$$\begin{aligned}
p - \bar{\theta}(0, k, p) &= \int_{p_{min}}^p dx - \int_{p_{min}}^p \frac{\partial \bar{\theta}}{\partial x}(0, k, x)dx \\
&= \int_{p_{min}}^p \left(1 - \frac{\partial \bar{\theta}}{\partial x}(0, k, x) \right) dx \geq 0 \quad (A.39)
\end{aligned}$$

since the integrand is positive for all values in the support of p . ■

A.4. CEO Piece Rate Derivation

Horizontal Hires. The bargaining condition (A.28) implies:

$$\begin{aligned}
r = & \beta_1 \left(p' + \lambda_1 \beta_1 \int_{p'}^{p_{max}} \phi_C(x)S(x)dx \right) + (1 - \beta_1) \left(p + k_- + \lambda_1 \beta_1 \int_{p+k_-}^{p_{max}} \phi_C(x)S(x)dx \right) \\
& - p' - \lambda_1 \beta_1 \int_{p'}^{p_{max}} \phi_C(x)S(x)dx - \lambda_1(1 - \beta_1) \int_{p+k_-}^{p'} \phi_C(x)S(x)dx \quad (A.40)
\end{aligned}$$

Combining terms yields:

$$\begin{aligned}
r &= -(1 - \beta_1)(p' - p - k_-) - \lambda_1(1 - \beta_1)^2 \int_{p+k_-}^{p'} \phi_C(x)S(x)dx \\
&= -(1 - \beta_1) \int_{p+k_-}^{p'} \frac{\rho + \mu + \eta + \lambda_1 S(x)}{\rho + \mu + \eta + \lambda_1 \beta_1 S(x)} dx \\
&= \int_{p+k_-}^{p'} q(x)dx \equiv r_C^{hor}(p', 0, p + k_-)
\end{aligned} \tag{A.41}$$

Diagonal Hires. Applying the definition of $\bar{\theta}$ (Equation (5)), the bargaining condition (A.26) can be rewritten as:

$$V_C(r, 0, p) = \beta_1 V_C(0, 0, p') + (1 - \beta_1) V_C(0, k, \bar{\theta}(0, k, p)) \tag{A.42}$$

Inserting the associated value functions then yields:

$$\begin{aligned}
r &= \beta_1 \left(p' + \lambda_1 \beta_1 \int_{p'}^{p_{max}} \phi_C(x)S(x)dx \right) + (1 - \beta_1) \left(\bar{\theta}(p, k_-) + k_- + \lambda_1 \beta_1 \int_{p+k_-}^{p_{max}} \phi_C(x)S(x)dx \right) \\
&\quad - p' - \lambda_1 \beta_1 \int_{p'}^{p_{max}} \phi_C(x)S(x)dx - \lambda_1(1 - \beta_1) \int_{\bar{\theta}(0, k_-, p) + k_-}^{p'} \phi_C(x)S(x)dx
\end{aligned} \tag{A.43}$$

Collecting terms and rearranging yields:

$$r = -(1 - \beta_1)(p' - \bar{\theta}(p, k_-) - k_-) - \lambda_1(1 - \beta_1)^2 \int_{\bar{\theta}(0, k_-, p) + k_-}^{p'} \phi_C(x)S(x)dx \tag{A.44}$$

$$= -(1 - \beta_1) \int_{\bar{\theta}(0, k_-, p) + k_-}^{p'} \frac{\rho + \mu + \eta + \lambda_1 S(x)}{\rho + \mu + \eta + \lambda_1 \beta_1 S(x)} dx \tag{A.45}$$

$$= -(1 - \beta_1) \int_{\bar{\theta}(0, k_-, p) + k_-}^{p'} q(x)dx \tag{A.46}$$

$$= r_C^{hor}(p', 0, p + k_-) - (1 - \beta_1) \int_{\bar{\theta}(0, k_-, p) + k_-}^{p+k_-} q(x)dx \equiv r_C^{diag}(p', 0, p + k_-) \tag{A.47}$$

Vertical Hires. Similar to the previous case, we can apply the definition of $\bar{\theta}$ and rewrite the bargaining condition (A.8) as:

$$V_C(r, k, p') = \beta_0 V_C(0, k, p') + (1 - \beta_0) V_C(0, k, \bar{\theta}(r, k, p)) \quad (\text{A.48})$$

Inserting the definition of the value function yields:

$$\begin{aligned} r = & \beta_0 \left(p' + k + \lambda_1 \beta_1 \int_{p'}^{p_{\max}} \phi_C(x) S(x) dx \right) + (1 - \beta_0) \left(\bar{\theta}(r, k, p) + k + \lambda_1 \beta_1 \int_{p+k_-}^{p_{\max}} \phi_C(x) S(x) dx \right) \\ & - p' - k - \lambda_1 \beta_1 \int_{p'}^{p_{\max}} \phi_C(x) S(x) dx - \lambda_1 (1 - \beta_1) \int_{\bar{\theta}(r, k, p) + k}^{p'} \phi_C(x) S(x) dx \end{aligned} \quad (\text{A.49})$$

Substituting the identity $\beta_0 = \beta_0 + \beta_1 - \beta_1$ and collecting terms yields:

$$\begin{aligned} r = & \overbrace{-(1 - \beta_1) \int_{\bar{\theta}(r, k, p) + k}^{p'} q(x) dx}^{r_C^{\text{diag}}(p', 0, p+k)} - (1 - \beta_1) \int_{p'}^{p'+k} q(x) dx \\ & + (\beta_0 - \beta_1) \int_{\bar{\theta}(r, k, p) + k}^{p'} \frac{\rho + \mu + \eta}{\rho + \mu + \eta + \lambda_1 \beta_1 \bar{F}(x)} dx \equiv r_C^{\text{vert}}(p', 0, p + k) \end{aligned} \quad (\text{A.50})$$

A.5. Executive Piece Rate Derivation

The bargaining condition for horizontal executive moves (Equation (A.16)) implies:

$$\begin{aligned} r = & \beta_1 \left(p' + \lambda_0 \beta_0 \int_{\bar{\theta}(p', 0)}^{p_{\max}} \phi_C(x) S(x) dx + \lambda_1 \beta_1 \int_{\bar{\theta}(p, k_-)}^{p_{\max}} \phi_C(x) S(x) dx + \lambda_2 \beta_1 \int_{p'}^{p_{\max}} \phi_E(x) S(x) dx \right) \\ & + (1 - \beta_1) \left(p + k_- + \lambda_0 \beta_0 \int_{\bar{\theta}(p, k_-)}^{p_{\max}} \phi_C(x) S(x) dx + \lambda_1 \beta_1 \int_{\bar{\theta}(p, k_-) + k_-}^{p_{\max}} \phi_C(x) S(x) dx + \lambda_2 \beta_1 \int_{p+k_-}^{p_{\max}} \phi_E(x) S(x) dx \right) \\ & - p' - \lambda_0 \beta_0 \int_{\bar{\theta}(p', 0)}^{p_{\max}} \phi_C(x) S(x) dx - \lambda_1 \beta_1 \int_{\bar{\theta}(p, k_-)}^{p_{\max}} \phi_C(x) S(x) dx - \lambda_2 \beta_1 \int_{p'}^{p_{\max}} \phi_E(x) S(x) dx \\ & - \lambda_1 (1 - \beta_1) \int_{\bar{\theta}(p, k_-) + k_-}^{\bar{\theta}(p', 0)} \phi_C(x) S(x) dx - \lambda_2 (1 - \beta_1) \int_{p+k_-}^{p'} \phi_E(x) S(x) dx \end{aligned} \quad (\text{A.51})$$

Simplifying and collecting terms yields:

$$\begin{aligned}
r = & -(1 - \beta_1)(p' - p - k_-) - \lambda_2(1 - \beta_1)^2 \int_{p+k_-}^{p'} \phi_E(x)S(x)dx \\
& - \lambda_1(1 - \beta_1)^2 \int_{\bar{\theta}(p,k_-)+k_-}^{\bar{\theta}(p',0)} \phi_C(x)S(x)dx + \lambda_0\beta_0(1 - \beta_1) \int_{\bar{\theta}(p,k_-)}^{\bar{\theta}(p',0)} \phi_C(x)S(x)dx
\end{aligned} \tag{A.52}$$

which is equivalent to Equation (20).

A.6. Characterization of the CEO Renegotiation Set

We restrict the analysis of this section to the special case of the model in which $\beta_0 = \beta_1$; more general results are a work in progress. The renegotiation set is defined as the closed interval $[\theta_C(p, k), p + k]$ with mass $\Omega_r^m(p, k)$ for $m \in \{hor, diag, vert\}$. $\theta_C(p, k)$ is defined implicitly by the following equation:

$$r = - \int_{\theta_C(p,k)}^{p+k} q(x)dx \tag{A.53}$$

where the left hand side denotes the CEO's current piece rate and the right hand side is the revised piece rate in case of a renegotiation. The form of the piece rate r , of course, depends on the path up the job ladder the manager took before landing in the CEO position. We begin with the case of horizontally-hired CEOs. Letting $p_- + k_-$ denote the sum of the match productivity and firm-specific human capital associated with the CEO's previous position at time of departure, inserting the definition of r_C^{hor} (Equation (14)), and rearranging terms yields:

$$\int_z^p q(x)dx = \int_{\theta_C(p,k)}^{p+k} q(x)dx \tag{A.54}$$

$$\int_p^{p+k} q(x)dx = \int_{p_-+k_-}^{\theta_C(p,k)} q(x)dx \tag{A.55}$$

Equation (A.55) implies that at time of hire (i.e. when $k = 0$), $\theta_C(p, 0) = p_- + k_-$. The dynamics of $\theta_C(p, k)$ as k increases can be obtained by implicitly differentiating (A.55) with respect to k :

$$\frac{\partial \theta_C}{\partial k}(p, k) = \frac{q(p+k)}{q(\theta_C(p, k))} < 1 \tag{A.56}$$

where the inequality follows from the fact that $q(x)$ is monotonically decreasing. Additionally, the differential equation (A.56) is independent of $p_- + k_-$. Thus, while the initial value of θ_C upon taking a new CEO position will differ by CEO type, the slope of θ_C with respect to k is independent of CEO type. Using the same approach as above, the initial condition of θ_C for diagonally-hired CEOs is apparent by inspecting the equation:

$$\int_p^{p+k} q(x)dx = \int_{\bar{\theta}(p_-, k_-) + k_-}^{\theta_C(p, k)} q(x)dx \quad (\text{A.57})$$

Hence, for diagonally-hired CEOs $\theta_C(p, 0) = \bar{\theta}(p_-, k_-) + k_-$. Characterizing θ_C for vertically-hired CEOs is made difficult by the fact that $\beta_0 \neq \beta_1$ in general. We thus restrict attention to the special case where $\beta_0 = \beta_1$ for now. Consider then a vertically-hired CEO with τ units of firm-specific tenure at time of hire. We can write:

$$\int_{\bar{\theta}(p_-, k) + k}^{p+k(\tau)} q(x)dx = \int_{\theta_C(p, k)}^{p+k(\tau+\Delta)} q(x)dx \quad (\text{A.58})$$

$$\int_{p+k(\tau)}^{p+k(\tau+\Delta)} q(x)dx = \int_{\bar{\theta}(p_-, k) + k}^{\theta_C(p, k)} q(x)dx \quad (\text{A.59})$$

At the point of promotion (when $\Delta = 0$), $\theta_C(p, k(\tau)) = \bar{\theta}(p_-, k) + k$. Because θ_C monotonically increases with k , this implies that the value of $\theta_C(p, 0)$ is smaller for vertically-hired CEOs relative to diagonally-hired CEOs. With these results in hand, we can state formally the proof of Theorem 2:

Proof of Theorem 2. Define $\Omega_r^m(p, k) = p + k - \theta_C(p, k)$. Differentiating with respect to p :

$$\frac{\partial \Omega_r^m}{\partial k}(p, k) = 1 - \frac{\partial \theta_C}{\partial k}(p, k) > 0 \quad (\text{A.60})$$

which follows from the fact that $\frac{\partial \theta_C}{\partial k}(p, k) < 1$. Let $z^m = \theta_C(p, 0)$ for $m \in \{hor, diag, vert\}$ denote the type-specific initial condition of $\theta_C(p, k)$. As $z^{vert} < z^{diag} < z^{hor}$, it follows that for any (p, k) :

$$\Omega_r^{vert}(p, k) > \Omega_r^{diag}(p, k) > \Omega_r^{hor}(p, k) \quad (\text{A.61})$$

■

We have

$$\begin{aligned} r_C^{vert}(p', 0, p' + k_-) &= r_C^{diag}(p', 0, p' + k_-) - (1 - \beta_1) \int_{p'}^{p' + k_-} q(x) dx \\ &\quad + (\beta_0 - \beta_1) \int_{\bar{\theta}(p) + k_-}^{p'} \frac{\rho + \mu + \eta}{\rho + \mu + \eta + \lambda_1 \beta_1 \bar{F}(x)} dx \end{aligned} \quad (\text{A.62})$$

The proof holds trivially if $\beta_0 = \beta_1$ or $\beta_1 = 1$. Suppose $\beta_0 \leq \beta_1$. Consider an offer to be internally promoted to CEO such that $p' = \bar{\theta}$, i.e. the threshold offer. Any other accepted offer will be better. Then, we need to show that

$$-(1 - \beta_1) \int_{\bar{\theta}}^{\bar{\theta} + k_-} q(x) dx + (\beta_1 - \beta_0) \int_{\bar{\theta}}^{\bar{\theta} + k_-} \frac{\rho + \mu + \eta}{\rho + \mu + \eta + \lambda_1 \beta_1 \bar{F}(x)} dx \leq 0$$

We can write

$$-(1 - \beta_1) \int_{\bar{\theta}}^{\bar{\theta} + k_-} q(x) dx + (\beta_1 - \beta_0) \int_{\bar{\theta}}^{\bar{\theta} + k_-} q(x) dx - (\beta_1 - \beta_0) \int_{\bar{\theta}}^{\bar{\theta} + k_-} \frac{\lambda_1 S(x)}{\rho + \mu + \eta + \lambda_1 \beta_1 \bar{F}(x)} dx$$

The third term is ≤ 0 , so we only need show

$$-(1 - \beta_1) \int_{\bar{\theta}}^{\bar{\theta} + k_-} q(x) dx + (\beta_1 - \beta_0) \int_{\bar{\theta}}^{\bar{\theta} + k_-} q(x) dx \leq 0$$

This holds surely if $\beta_1 \leq \frac{\beta_0 + 1}{2}$. So, we have the statement:

If $\beta_0 \leq \beta_1 \leq \frac{\beta_0 + 1}{2}$ then vertically hired are always weakly smaller contracts. This holds for all values $\{(\beta_0, \beta_1) \in [0, 0.5]^2 \text{ s.t. } \beta_0 \leq \beta_1\}$.

However, the sharp bound is

$$(\beta_1 - \beta_0)[1 - A(r, p, k_-)] \leq (1 - \beta_1)$$

where

$$A(r, p, k_-) = \frac{\int_{\hat{\theta}(r, p, k_-)}^{\hat{\theta}(r, p, k_-) + k_-} v(x) dx}{\int_{\hat{\theta}(r, p, k_-)}^{\hat{\theta}(r, p, k_-) + k_-} q(x) dx} \leq 1$$

$$\text{and } v(x) = \lambda_1 S(X) / (\rho + \mu + \eta + \lambda_1 \beta_1 S(x))$$

B. Estimation Appendix

B.1. Weighting Matrix

From the empirical sample we obtain a $K \times 1$ vector of moments $\hat{M}$. Let Ψ denote the corresponding $N \times K$ matrix of influence functions, N being the number of observations in the sample. Each element Ψ_{nk} is the influence function describing observation n 's contribution to moment k . The covariance matrix of the vector of moments can then be estimated as:

$$a\hat{v}ar(\hat{M}) = \Psi' \Psi \quad (\text{B.1})$$

The weighting matrix $\hat{W}$ is then obtained as the inverse of matrix [B.1](#). Let $\Theta \in \mathbb{R}^P$ denote an arbitrary vector of structural parameters. Define the moment residual $g : \mathbb{R}^P \rightarrow \mathbb{R}^M$ as:

$$g(\Theta) = \hat{M} - \frac{1}{S} \sum_{s=1}^S \hat{m}^s(\Theta) \quad (\text{B.2})$$

Where $\hat{M}$ is the vector of empirical moments, $\hat{m}^s(\Theta)$ is the vector of simulated moments given parameter values Θ in simulation s , and S is the total number of simulations. The vector of estimates $\hat{\Theta}$ minimizes the SMM objective function:

$$\hat{\Theta} = \arg \min_{\Theta} g(\Theta) \hat{W} g(\Theta)' \quad (\text{B.3})$$

B.2. Model Estimation Algorithm

We minimize the SMM quadratic form using the particle swarm algorithm. The routine proceeds as follows:

1. *Set initial guesses for model parameters:* We set initial values for the structural parameters Θ . The initial guess is chosen manually, while subsequent guesses are selected by the particle swarm algorithm.
2. *Simulate model:*
3. *Construct simulated panel and compute moments:* Using the simulated data, we construct a panel resembling the empirical sample and compute the same moments as described in Section 4.
4. *Evaluate objective function:* Given the set of simulated moments, we evaluate the SMM objective function (). If the objective function value satisfies the particle swarm stopping criterion, the algorithm halts. Otherwise, a new candidate parameter vector Θ' is selected and steps 2-5 repeat. This continues until the algorithm halts.